

Master Thesis

He's the Greatest Dancer: Detecting Linguistic Gender from Explicit Cues

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Abstract

The gender of a person referred to in English text is usually signalled through explicit referential cues, such as pronouns, gendered titles, or lexical descriptions, rather than encoded grammatically. However, languages that have grammatical gender, such as German, require gender to be specified for most nouns, especially those referring to a person. This poses a problem for machine translation, where implicit gender associations in the training data can be reflected and amplified whenever the source language carries less gender information than the target language demands. Gender-fair language provides ungendered forms of typically gendered nouns to avoid gender assumptions in cases where gender is unknown or should not be specified. However, it is usually underrepresented in commonly used training data and therefore rarely utilised by MT models. Although LLMs can generate gender-fair forms, they struggle to identify when to apply them. Systems that annotate linguistic gender from explicit cues in the English source can therefore serve as a preprocessing step that identifies which translation is appropriate. This thesis investigates how well the linguistic gender of nouns referring to people in English text passages can be detected by systems that only have access to explicit gender cues instead of the full context, based on the hypothesis that restricting a system in this way might help avoid misclassifications caused by gender stereotypes. Rule- and feature-based classifiers limited to explicit cues were compared with DistilBERT and Qwen 2.5 in its 3B and 72B variants, which had access to the full context. The models were evaluated on weighted F1-score and on the extent to which their misclassifications aligned with common gender stereotypes. The systems with access to the full context achieved higher weighted F1-scores for cases where gender was indicated through pronouns, gendered titles, or lexical references. The explicit-cue classifiers performed better for nouns whose gender was indicated through lexical gender. Qwen 2.5-3B showed the lowest performance across all cue types. In cases where no explicit gender cue was present, DistilBERT's misclassifications largely aligned with female gender stereotypes and Qwen 2.5-3B's with female and male stereotypes, while the explicit-cue classifiers and Qwen 2.5-72B showed no noticeable tendency towards harmful gender bias. Explicit-cue systems therefore offer a less resource-intensive alternative to larger decoders and a less gender-biased alternative to DistilBERT for avoiding misclassifications caused by gender stereotypes.

The code and datasets used are available at
`https://github.com/dasmarie/linguistic-gender-classification`

Declaration of Authorship

I, Marie Necker, declare that this thesis, titled *He's the Greatest Dancer: Detecting Linguistic Gender from Explicit Cues* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: August 16, 2026

Signed: 

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Chapter 1

Introduction

Between different languages, the extent to which gender is specified varies. While some languages, such as French or German, encode gender grammatically, others, such as English, mostly express gender through lexical references, and others, such as Turkish, barely encode gender at all (Corbett, 2013a; Eisenberg, 2013). In the English language, it is customary to indicate an entity's gender by using explicit referential cues, such as pronouns, titles, terms containing lexical gender, or other lexical descriptions (Corbett, 1991). Therefore, if a noun itself is not gendered and no referential cues, such as pronouns, indicate otherwise, its gender is generally unknown. However, due to historically established gender stereotypes, certain attributes or nouns, such as *nurse*, are often associated with a certain gender (Williams and Best, 1990), which can lead to harmful and stereotype-reproducing assumptions in contexts where gender is not specified. These implicit gender associations are especially relevant to the field of language technology, since training data often contains gender stereotypes present in society, which are then reflected and amplified by language models (Bolukbasi et al., 2016; Caliskan et al., 2016; Kotek et al., 2023). This thesis therefore aims to test how well the linguistic gender of nouns referring to people in English text passages can be detected by systems that only get access to explicit gender cues instead of the full context.

1.1 Relevance

While systems classifying nouns by linguistic gender could be useful for several tasks that are affected by harmful gender biases, especially in the form of gender bias evaluation, the relevance of such a system will be explained based on automatic translation from English to German.

Since German contains grammatical gender (Eisenberg, 2013) while English, arguably, does not (Corbett, 1991), gender often needs to be explicitly specified in German text where it is not specified in English. To name an example, traditionally, 'the teacher' would have to either be translated as 'die Lehrerin' (feminine) or 'der Lehrer' (masculine). Although there are several approaches that provide gender-neutral versions of typically gendered words, such as 'die Lehrperson' (neutral), they have not been applied long and extensively enough to be properly represented in the training data of MT systems. Instead, generic masculine forms and gender stereotypes are prevalent (Savoldi et al., 2025). As a result, gender-fair language is hardly present in the output, and the translations of binary-gender nouns often align with gender stereotypes. Im-

plicit gender cues tied to stereotypes can contribute to the mistranslation of a word's gender, even when it is explicitly indicated (p. 8).

While LLMs are generally successful at producing gender-neutral forms in translation, they fail to correctly identify when to use them, which results in them heavily over- or underusing gender-fair language (Pranav et al., 2025). Systems that annotate linguistic gender based on explicit gender cues, like the systems investigated in this thesis, could therefore potentially offer a preprocessing step that might help solve this issue by identifying the gender ascribed to an entity in English source text to find out which translation would be appropriate.

1.2 Problem Definition

Although language models trained on large amounts of human text are generally good at picking up on implicit cues that signal gender (Ananya et al., 2019), such as attributes or occupation titles Williams and Best (1990), these implicit cues are often connected to stereotypes that are reinforced when learned by language technology. This thesis therefore aims to provide a system that identifies linguistic gender from explicit cues, such as pronouns, lexical gender, and lexical descriptions, to provide information on the gender ascribed to an entity in a given text while avoiding classifications due to stereotypes that are implicitly encoded in language.

To label the linguistic gender of the English target nouns in their dataset for gender-fair machine translation from English to German, Pranav et al. (2025) employed different language models. Out of those models, Qwen 2.5-72B (Yang et al., 2024) showed the significantly highest performance, with a weighted F1 of 0.78. This model is therefore going to be used as a baseline.

1.3 Research Goal

The primary goal of this thesis is to find out how well a system that specifically targets only explicit gender cues, using coreference resolution, word definitions, and dependency relations, performs at detecting referential gender compared to an LLM, specifically Qwen 2.5 72B (Yang et al., 2024), which reached the best weighted F1-score out of the large language models employed by Pranav et al. (2025) to label the Glitter dataset.

Next to comparing weighted F1 scores achieved by the different systems, this thesis will also focus on whether and to what extent misclassifications of the applied systems align with common gender stereotypes.

1.4 Approach

General Task Definition: All systems applied in this thesis are provided with text passages that contain a target noun whose linguistic gender needs to be classified. Rather than detecting the real-world gender of an entity, the systems should detect the linguistic gender that can be deduced from a given text passage. If the gender of a noun is not specified, it should be classified as `ambiguous`. Otherwise, the noun's linguistic gender should be identified as `unambiguous_female`, `unambiguous_male`, or `unambiguous_all` for mixed-gender groups. A more detailed description of the task can be found in Chapter 4.

To find out how systems limited to explicit gender cues perform at detecting linguistic gender compared to systems with access to the full context, six different system variations are tested and compared, regarding weighted F1-scores and stereotypical misclassifications.

- *Rule-Based Classifier*: This system applies a pipeline that utilises coreference resolution, WordNet glosses, and pre-defined dependency relations between explicit gender cues and the target noun.

- *Feature-Based Classifier*: A logistic regression model is trained on features derived from the components of the rule-based classifier.

- *Rule-Based Classifier + Qwen 2.5-3B*: First, the rule-based system classifies all relevant nouns. Then, for all instances that were classified as `ambiguous`, Qwen 2.5-3B is provided with gender cues from the surrounding text passages, which it is then asked to assign to the correct noun.

- *Qwen 2.5-3B*: To compare how smaller decoders deal with identifying a noun's linguistic gender and connected stereotypes compared to larger ones, a smaller large language model is applied to the task. Both LLMs are provided with the same modified few-shot prompt based on the prompt used by Pranav et al. (2025) to annotate the Glitter dataset.

- *Qwen 2.5-72B*: This specific large language model performed best at labeling referential gender for Pranav et al. (2025). It is therefore used as a comparison to the systems based on explicit cues.

- *DistilBERT*: DistilBERT (Devlin et al., 2019; Sanh et al., 2020) is fine-tuned for classification with access to the full context of each example.

To cover different types of explicit cues, such as pronouns, lexical gender, gendered titles, and lexical references, the dataset consists of multiple components, which are explained in Chapters 3 and 4.

1.5 Results

The results suggest that systems detecting gender from explicit cues can be successful at classifying nouns by linguistic gender whenever clear gender cues, such as pronouns, titles, or lexical gender are provided. In contexts where gender needs to be deduced from lexical descriptions and contextual cues, their performance is, however, limited. While the rule-based system does not outperform Qwen 2.5-72B, it clearly outperforms the smaller LLM, Qwen 2.5-3B. Although DistilBERT performs best for most types of gender cues, its performance cannot be taken at face value due to constraints in the dataset. Misclassifications due to gender stereotypes cannot be observed for the rule- and feature-based classifiers and Qwen 2.5-72B. However, in contexts where no explicit gender cues are provided, DisitlBERT's misclassifications align with female gender stereotypes while Qwen 2.5-3B's misclassifications partly align with male and female gender stereotypes.

1.6 Disclaimer on Gender Labels

Although gender is made up of several factors and should be viewed as a spectrum (Bing and Bergvall, 1996; van Anders et al., 2017), this thesis works with categories that are necessary for the classification task and that try to encapsulate how gender is expressed through language. While, in the English language, linguistic gender is shaped by the real-world gender of the people it refers to, it often fails to capture its complexity (Bucholtz, 1999). The categories used in this thesis therefore work with categories of which one represents female gender, one represents male gender, and one represents gender identities that are unspecified or fall outside the gender-binary.

Chapter 2

Literature Review

This chapter places this thesis's topic within existing work on linguistic gender and language technology. It provides an overview of how gender can be indicated in language, how it is sometimes falsely inferred through stereotypical associations, and how those associations can cause harmful gender bias in language technology.

The chapter's main focus is to establish the difference between explicitly marked linguistic gender and textual elements, such as attributes or occupation titles, which do not mark gender but are often associated with a certain gender. To show how these associations can be relevant to language technology, the chapter uses automatic translation from English to German as an example. This example illustrates why classifying gender from explicit gender cues, such as pronouns, lexical gender, or lexical descriptions instead of the full context, could help avoid errors caused by gender stereotypes.

2.1 Gender in Language

The following chapter examines how gender is expressed and understood through language. To achieve this, it explores explicit linguistic cues and implicit gender associations, including language strategies that are used to make language less biased towards gender stereotypes.

2.1.1 Definitions and Categories

To discuss gender in a linguistic context, it is useful to first define the concepts the term encompasses, how they differ, and how they might influence each other. Since gender is a widely discussed topic that spans different disciplines, its definition and classification can vary. To keep the definitions comprehensive, they will mainly be based on a gender framework proposed by Cao and Daumé III (2020).

To understand how gender operates in language, many draw a first distinction between sociological and linguistic gender (Ackerman, 2019; Cao and Daumé III, 2020). While sociological or (bio-)social gender is often described as including one's own experience of gender, its expression, and how expectations regarding established gender stereotypes might influence both (Ackerman, 2019), linguistic gender describes how gender is represented in language, sometimes trying to capture social gender (Hellinger and Bußmann, 2001).

Inspired by earlier works (Corbett, 1991, 2013b; Fuertes-Olivera, 2007; Ackerman, 2019) on gender and language, Cao and Daumé III (2020) split linguistic gender into

grammatical, referential, and lexical gender.

Grammatical gender generally describes a system in which, in certain languages, nouns can be grouped into different genders according to grammatical properties and agreement (Ackerman, 2019), which can be realised in different ways. While in some languages the grammatical properties mentioned can reflect social gender, in other languages this connection is not established (Cao and Daumé III, 2020), and yet others do not have grammatical gender at all.

An example of a language that uses grammatical gender is German, in which nouns are categorised into three gender classes: feminine, masculine, and neuter (Auer and Siegel, 2021). While nouns referring to entities with biosocial gender, primarily humans, often show an alignment between grammatical and social gender, nouns without such referents are assigned grammatical gender in a way that, while sometimes morphologically or semantically motivated (Eisenberg, 2013), is, to some level, arbitrary and cannot be reliably inferred from semantic properties alone (Corbett, 1991; Comrie, 1999).

Referential gender, on the other hand, is relevant only for entities linked to real-world gender, which can be expressed through markers such as pronouns or gendered nouns referring to the entity. The gender of these nouns is defined by lexical gender, which Cao and Daumé III (2020) define as the gender inherent to a certain linguistic unit, such as 'monk' carrying masculine gender, while the word 'nun' would typically be identified as feminine and 'painter' would be seen as neutral.

English represents an example of a language that largely relies on referential and lexical gender and is often characterised as having a gender system that is based on semantic criteria (Corbett, 1991). This implies that the gender of a noun is typically determined by the gender identity of the real-world entity to which it refers. English, therefore, does not grammatically classify all nouns by gender (Hellinger and Bußmann, 2001), but it assigns gender through references such as pronouns or morphemes that refer to an entity's social gender (Curzan, 2003).

However, although social gender heavily influences how gender is represented in language, it is important to note that linguistic gender does not necessarily reflect a person's real-world gender, as it often fails to capture its complexity (Bucholtz, 1999).

2.1.2 Implicit Gender Associations

While linguistic gender demonstrates how language encodes gender explicitly, stereotypes that have been established historically (Gnadt et al., 2025) and cross-culturally (Williams and Best, 1990) are reflected in language (Beukeboom, 2025; Menegatti and Rubini, 2017) and can lead to implicit associations between certain linguistic elements and gender. This correlation can manifest in men being assigned traits such as rationality, strength, and ambition, whereas women are often portrayed as emotional and passive while also being defined in relation to their appearance and attractiveness (Williams and Best, 1977, 1990; Hoyle et al., 2019). Furthermore, while terms usually associated with women tend to align with a rather narrow and specific stereotype, the pool of terms associated with male gender seems to be larger and more general (Kotek et al., 2023). Consequently, masculine gender tends to be the default association in most contexts, whereas feminine gender is mostly inferred in more specific, stereotypically feminine contexts. Non-binary gender, however, is, in most cases, not inferred from implicit associations only and needs explicit cues, such as neo-pronouns, to be elicited from text (Renström et al., 2023).

It has furthermore been shown that grammatical gender can influence how people subconsciously assign social gender to words, even with non-human entities (Casado et al., 2023) and that the human brain can process information conforming to known gender stereotypes more easily (Wang et al., 2017). Correspondingly, certain occupation titles that have historically been associated with one gender are often automatically perceived in relation to that gender, which is maintained and reinforced through implicitly gendered wording used in job advertisements(Gaucher et al., 2011).

As a result, gender is often inferred, even if not explicitly stated (Ananya et al., 2019). The fact that these linguistic patterns can help gender stereotypes persist, which can be perpetuated further through language technology (Bender et al., 2021) based on word embeddings (Zhao et al., 2019), underlines the importance of gender-neutral expressions in contexts where gender is not given through pronouns or other explicit cues.

2.1.3 Gender-Fair Language

In English, most nouns are gender-neutral on their own, since gender is mostly indicated through referential cues. Languages with grammatical gender, such as German, by contrast, often demand gender information for nouns referring to people. This difference can cause mistranslations caused by gender stereotypes and implicit gender associations in automatic and manual translation from English to German, as well as in other language pairs where the source language provides less gender information than the target language demands. This issue is explained in Section 2.2.2.

Since forcing all nouns into binary gender categories requires knowing their referents' gender and excludes gender identities outside this binary system, several approaches have been developed within these languages to avoid such assumptions. German, for example, currently exhibits four major methods for avoiding binary gender references: gender-neutral rewording, gender-inclusive characters, gender-neutral endings, and neosystems (Lardelli and Gromann, 2023a).

Gender-neutral rewording represents one of the most straight-forward and widely accepted strategies for gender-fair language. To avoid assuming gender, nouns can be replaced by neutral nouns, combined with neutral endings or nouns such as -kraft or -person, or omitted through infinitive and passive constructions (Lardelli and Gromann, 2023a).

A widely used (Dick et al., 2024) strategy intended to include all genders in a single expression lies in the use of *gender-inclusive characters*, where the masculine or base form of a word is separated from the feminine suffix to make both forms visible.

Furthermore, *gender-neutral characters or endings* can be attached to the word stem to create new, gender-neutral word forms, which can also be applied as pronouns if used by themselves (Hornscheidt and Sammla, 2021).

Neosystems, however represent new grammatical systems that invent a fourth gender to give non-binary people more visibility (Lardelli and Gromann, 2023a).

Since these approaches have only been used more extensively in the last few years, they are often underrepresented in the data used to train MT systems, which is why they are often underused (Lardelli and Gromann, 2023b). The main issue with gender-fair translation does therefore not lie in the lack of gender-neutral alternatives but in their correct application. This will, however, be explained in more detail in Section 2.2.2.

Gender-Neutral Rewording	Avoids marking gender by using or creating compounds with neutral nouns	Lehrkraft (teaching staff), Lehrperson (teaching person), Lehrende (teachers, present participle of 'teach')
Gender-Inclusive Characters	Uses typographical symbols to separate word stems from feminine endings, intended to include all genders beyond the binary	Lehrer*in (gender star), Lehrer_in (underscore), Lehrer:in (colon), Lehrer/in (slash)
Gender-Neutral Endings	Replacing binary endings with neutral characters (like x) or new endings (like -ens) derived from the word Mensch	Lehrens (-ens), Lehrx (-x), einx schlaux Lehrx (a smart teacher, -x)
Neosystems	New grammar systems, introducing new and unique sets of pronouns and articles	NoNa system (hen), xier system (xier)

Table 2.1: Selected Gender-Fair Language Strategies for German based on Lardelli and Gromann (2023a)

2.2 Gender Bias in Language Technologies

Human gender stereotypes that are reflected in language (Williams and Best, 1977; Mills, 2008; Gaucher et al., 2011; Menegatti and Rubini, 2017), are mirrored and amplified in static (Bolukbasi et al., 2016; Caliskan et al., 2016) and contextualised word embeddings (Zhao et al., 2019; Bartl et al., 2020), and can consequently also be detected in language technologies such as machine translation systems (Savoldi et al., 2025) or large language models (Sheng et al., 2019; Gnadt et al., 2025). Although implicit contextual cues can be helpful in predicting gender (Ananya et al., 2019), they can be misleading in non-stereotypical contexts and are likely unhelpful for entities outside of the gender-binary, such as non-binary individuals or mixed groups.

This section starts out by exploring how gender bias manifests in word embeddings, which most modern language technologies are based on, to then explain how these stereotypes influence downstream applications, which can reproduce these stereotypes and therefore potentially slow down social progress (Bender et al., 2021).

2.2.1 Word Embeddings

Textual information in the form of, for instance, attributes, themes, or occupations (Kotek et al., 2023) can carry implicit gender signals that manifest in word embeddings. Caliskan et al. (2016) showed that gender stereotypes are encoded in static word embeddings by demonstrating correlations such as that career-related concepts cluster closer to male than female attribute words and that word embeddings can predict real-world distributions such as the share of women across the fifty most common professions in the United States. It was furthermore found that word embeddings capture stereotypical analogies, such as 'man' being to 'computer programmer' what 'woman' is to 'homemaker', and 'he' being to 'doctor' what 'she' is to 'nurse' (Bolukbasi et al.,

2016) and that gender is encoded geometrically as a specific direction within a word embedding's vector space.

Similar findings have been established for contextualised word embeddings, where it has been shown that word embeddings from ELMo and BERT inherit significant gender bias from training corpora in which male entities are frequently overrepresented (Zhao et al., 2019; Bartl et al., 2020). While gender might be less directly encoded in contextualised compared to static word embeddings (Basta et al., 2019), they still encode gender stereotypes in a lower-dimensional subspace, where male gender is encoded differently to female gender and can, likely due to its dominance in the training data, be predicted more easily than female gender (Zhao et al., 2019).

To mitigate these biases, several strategies have been developed. While it has been tried to remove the gender component from word embeddings (Bolukbasi et al., 2016), some argue that methods specifically targeting the gender direction only hide the bias but do not remove it, since language systems might also rely on other cues that might carry gender bias (Gonen and Goldberg, 2019). For downstream applications, possible methods include creating a gender-swapped version of the training data and retraining the model on a combined dataset of the original and the edited data, or creating a gender-swapped version of the test data and then averaging the embeddings with those of the original sentences to create neutralised representations (Zhao et al., 2019; Basta et al., 2019). While these approaches helped reduce bias in downstream applications, such as coreference resolution, they might prove difficult for morphologically richer languages, such as German, (Bartl et al., 2020), while also only operating within a binary gender system.

While binary gender representations are often defined by stereotypical associations, non-binary gender references tend to be erased due to being underrepresented in the textual data used to train language technology Dev et al. (2021), which can result in issues such as neo-pronouns not being recognised (p. 1970) while the singular 'they' tends to be less distinguishable from its plural form than binary pronouns (p. 1975).

2.2.2 Translation Systems

For machine translation, gender stereotypes might lead to mistranslations in contexts where the referential gender of an entity might not align with learned implicit gender cues in its context (Savoldi et al., 2025). Whenever the source language is less gendered than the target language, implicit gender cues tied to stereotypes can wrongly determine the gender of a translated word, even when an entity's gender is explicitly indicated (p. 8). Furthermore, when an entity's gender cannot be deduced from its textual environment, or whenever an entity does not fit into the gender binary, machine translation systems tend to default to the masculine form instead of using neutral or inclusive language (Pranav et al., 2025, p. 18455). However, since the pool of terms associated with male gender often tends to be larger and more general than the collection of terms associated with female gender, which tends to align with specific common stereotypes and is therefore also smaller (Kotek et al., 2023), the systems' defaulting to male translations holds true for most general terms while female translations mostly appear in contexts that fit the established stereotype. Meanwhile, gender-neutral, gender-inclusive, and non-binary language is hardly used (Lardelli and Gromann, 2023b; Pranav et al., 2025).

To mitigate these biases in machine translation, approaches such as data debiasing, fine-tuning (Tomalin et al., 2021), and gender-tagging (Stafanovičs et al., 2020;

Soler Uguet et al., 2023), amongst others, have been proposed. By annotating or tagging gender in the training data of MT systems, Stafanovičs et al. (2020) could generally improve translatin results. To annotate training examples, entities in the target language, containing grammatical gender, were tagged first, after which the tags were transferred to the source language, only containing notional gender (p.631). To test the system, which has to be done without a parallel target sentence, they used coreference resolution to detect pronouns in the input to help determine the entity's referential gender (p.632-633). A similar approach compared coreference resolution and a GPT-3.5 model to annotate the input. Better results were achieved with coreference resolution (Soler Uguet et al., 2023). However, in their test data, gender was only indicated through pronouns.

Generally, in context-aware machine translation, coreference resolution has successfully been used to improve translation quality, for instance by integrating it directly into the model (Ohtani et al., 2019; Lei et al., 2022; Vu et al., 2024) or, as mentioned above, by using it to construct augmented training data (Hwang et al., 2021).

Using large language models for automatic translation has shown that while they outperform traditional MT systems in handling anti-stereotypical information and perform better with neo-pronouns, they still often apply the generic masculinum (Savoldi et al., 2025). When prompted to provide gender-fair translations, they struggle to identify where to apply gender-fair language, resulting in them either heavily under- or overusing gender-fair expressions (Pranav et al., 2025). A system that identifies linguistic gender from explicit cues, such as pronouns or lexical references, might therefore help indicate to an LLM where gender-fair forms are appropriate and where they are not.

2.2.3 Linguistic Gender Resolution/Identification

Much of the work that is to some degree related to referential gender resolution has focused on gendered pronoun resolution as a coreference task, motivated by concerns about gender bias, resulting in benchmarks such as Winogender (Rudinger et al., 2018), WinoBias (Zhao et al., 2018), and GAP (Webster et al., 2018).

The gender-tagging methods used for gender-fair machine translation mentioned in Section 2.2.2, also only rely on pronouns or grammatical gender information from the target language (Stafanovičs et al., 2020; Soler Uguet et al., 2023).

Bartl and Leavy (2022) have presented a system to identify lexical gender from dictionary definitions.

The rule- and feature-based systems proposed in this thesis seek to combine the mentioned components, which are coreference resolution and lexical gender classification, and add pre-defined dependency paths to find referential cues that cannot be detected through a coreference system. In contrast to previous papers that include components on linguistic gender tagging, such as Soler Uguet et al. (2023) or Stafanovičs et al. (2020), this thesis goes beyond gendered pronouns as markers of gender and also includes lexical references, gendered titles, and lexical gender as possible gender cues to reflect the diversity of how gender can be expressed in the English language. The full method is described in Chapter 4.

Chapter 3

Data

To represent the variety of manners in which gender can be indicated in English text, a dataset that tests linguistic gender classification needs to contain varied passages that represent this diversity. As described in Section 2.1.1 and depicted in 4.1, in English, gender is usually indicated through referential gender cues or gender inherent to the word itself, also known as lexical gender.

With the dataset used in this thesis, I aim to cover instances of referential and lexical gender, which can include gender cues such as pronouns, lexical descriptions, titles, or a noun's lexical gender. The dataset used in this thesis consists of four components, of which three are pre-existing datasets from the field of gender-fair machine translation while one is an extension that I created to cover nouns containing lexical gender.

All components will be described according to the guidelines for data statements introduced by Bender and Friedman (2018). For the dataset extension, the extraction process will be described in more detail.

A more detailed description of the different types of gender cues relevant to this thesis can be found in Chapter 4.

Label	Dev (n=637)	Train (n=2754)	Test (n=1595)	Overall (n=4986)
ambiguous	221 (34.7%)	1043 (37.9%)	651 (40.8%)	1915 (38.4%)
unambiguous_all	7 (1.1%)	46 (1.7%)	32 (2.0%)	85 (1.7%)
unambiguous_female	180 (28.3%)	828 (30.1%)	464 (29.1%)	1472 (29.5%)
unambiguous_male	229 (35.9%)	837 (30.4%)	448 (28.1%)	1514 (30.4%)

Table 3.1: Label distribution per split

3.1 Overall

The full combined dataset contains 4986 text passages, of which each contains one target noun whose linguistic gender is labeled. The labels used are `ambiguous` for all instances where gender is not specified through the context or lexical gender, `unambiguous_female` for nouns with female gender, `unambiguous_male` for nouns with male gender, and `unambiguous_all` for plural nouns referring to groups that include different genders.

Overall, the test set contains slightly more instances of ambiguous (38.4%) target nouns, which therefore represent the majority class, while the amounts of male (30.4%) and female (29.5%) target nouns are relatively balanced. However, `unambiguous_all`

is heavily underrepresented with only 1.7% of all instances, which can be traced back to the fact that only the Glitter (Pranav et al., 2025) dataset works with plural nouns and that `unambiguous_all` is already heavily underrepresented there.

The dataset is split into a training, test, and development set, of which 50% form the training set, 10% are taken as a development set and the remaining 40% form the test set. The instances included in each split were chosen randomly by shuffling each dataset separately, with a seed of 14, before they were split into development, training, and test data.

Table 3.2 shows an overview of the different dataset components, which will be described in more detail in the following sections.

Table 3.2: Overview of the four evaluation datasets.

	FairTranslate	**GeNTE**	**Glitter**	**Lexical**
Size	1799	805	980	1402
Form	Single and multi-sentence passages (template)	Single sentences	Multi-sentence passages	Single-sentence excerpts
Phenomena	Pronouns	Pronouns, Titles, Lex. References	Lexical References	Lexical Gender
Source	Synthetic	EU parliament proceedings	Wikipedia and Synthetic	Gutenberg, Brown, Inaugural
Reference	Jourdan et al. (2025)	Piergentili et al. (2023)	Pranav et al. (2025)	self-created

3.2 FairTranslate

FairTranslate (Jourdan et al., 2025) was created to evaluate the ability of LLMs to provide gender-fair translations from English to French. It consists entirely of synthetic text. Each text passage in the dataset contains one of 62 seed nouns which are occupation titles that, based on real-world gender distribution data, were split into female-stereotyped, male-stereotyped, and gender-balanced nouns. The synthetic data was generated using GPT-4o and GPT o-1 under close human supervision to ensure variety, diversity, and correctness. They were then post-edited by the research team. All examples were generated in French first and translated to English after.

The full dataset consists of 2408 sentences of which each is annotated with the gender, ambiguity and stereotypical gender association of the seed noun. The seed noun itself is provided in French.

The examples consist of single- to multi-sentence statements of which each describes the actions or habits of a person with a certain occupation. While the writing style stays relatively simple, it uses a variety of different tenses. The gender cues (pronouns) are in various different positions.

In FairTranslate, gender is indicated through pronouns. It is therefore included to test a system's ability to deduce gender from related pronouns.

Dataset	Label	Overall	Dev	Train	Test
fairtranslate	ambiguous	808 (44.9%)	75 (41.9%)	474 (43.9%)	259 (48.0%)
	unambiguous_female	495 (27.5%)	50 (27.9%)	306 (28.3%)	139 (25.7%)
	unambiguous_male	496 (27.6%)	54 (30.2%)	300 (27.8%)	142 (26.3%)
gente	ambiguous	191 (23.7%)	19 (23.8%)	112 (23.2%)	60 (24.8%)
	unambiguous_female	307 (38.1%)	26 (32.5%)	187 (38.7%)	94 (38.8%)
	unambiguous_male	307 (38.1%)	35 (43.8%)	184 (38.1%)	88 (36.4%)
glitter	ambiguous	501 (51.1%)	46 (46.9%)	250 (51.0%)	205 (52.3%)
	unambiguous_all	85 (8.7%)	7 (7.1%)	46 (9.4%)	32 (8.2%)
	unambiguous_female	183 (18.7%)	17 (17.3%)	87 (17.8%)	79 (20.2%)
	unambiguous_male	211 (21.5%)	28 (28.6%)	107 (21.8%)	76 (19.4%)
lexical	ambiguous	415 (29.6%)	81 (28.9%)	207 (29.5%)	127 (30.2%)
	unambiguous_female	487 (34.7%)	87 (31.1%)	248 (35.4%)	152 (36.1%)
	unambiguous_male	500 (35.7%)	112 (40.0%)	246 (35.1%)	142 (33.7%)

Table 3.3: Label distribution per dataset per split.

3.3 GeNTE

GeNTE (Piergentili et al., 2023) is a test set that was designed to evaluate a model's ability to perform gender-fair translation from English to Italian. The entire dataset is sourced from the Europarl (Koehn, 2005) corpus which was chosen due to the high number of gender-ambiguous examples that it contains and because it provides formal text from the administrative domain where gender-fair translation is especially relevant.

Europarl (Koehn, 2005) is sourced from proceedings of the European Parliament between 1996 and 2005. Its English passages consist of sentences that are English in their original form and sentences that were translated from another European language. The English used can therefore include L1 and L2 English. Speakers can include politicians, officials, and translators of the European Union or anyone who might have spoken at the European parliament between 1996 and 2005.

Each example in the dataset consists of only one sentence. The language variety used can be described as formal administrative English. Most of the dataset is spoken English that was documented in written text.

The full dataset includes 1500 sentence pairs, each of which is annotated with the gender of all nouns referring to persons found in the sentence. Each sentence was edited by the research team to make sure it only contained referents of the same linguistic gender. For male gender, they differentiated between male pronouns that were used generically, which were labeled as neutral, and male pronouns that actually indicated male gender, which were labeled as male.

GeNTE uses a mix of different gender cues to indicate gender and is therefore the most representative of a system's ability to detect linguistic gender. It is furthermore the only dataset that uses titles as a gender cue.

3.4 Glitter

Glitter (Pranav et al., 2025) is a benchmark dataset that was created to evaluate gender-fair machine translation from English to German. It aims to represent a variety of different scenarios in which linguistic gender can be disambiguated. To achieve this, it uses text from the English Wikipedia as natural data and augmented the dataset with synthetic text to make it more balanced and diverse. The synthetic data was generated using using GPT-4o, after which it was post-edited by members of the research team.

The authors identify the English used in Wikipedia articles as Standard American English. The synthetic text is supposed to be in a similar style to these articles. The community of Wikipedia editors is large and diverse but consists of a majority of white men. The synthetic data was generated using using GPT-4o, after which it was post-edited by members of the research team. The dataset was annotated by professional translators who were German native speakers and had experience with gender-fair translation practices.

Glitter contains 980 passages based on 147 different gender-ambiguous seed-nouns in their plural form. Each passage consists of four sentences where one sentence containing the seed noun is preceded by two sentences and followed by one sentence. All sentences can but do not always contain gender cues which determine the gender of the seed noun. The gender of the seed noun is provided and labeled as either ambiguous, unambiguous_female, unambiguous_male, or unambiguous_all. The original dataset furthermore offers German gender-fair translations of the English text passages, which are, however, not used in this thesis.

Glitter only uses ambiguous plural seed nouns, which eliminates pronouns as a possible gender cue and therefore tests a system's ability to detect linguistic gender from lexical descriptions, coreferents and contextual cues.

Stereotype	Overall	Dev	Train	Test
none	3557 (71.3%)	483 (75.8%)	1907 (69.2%)	1167 (73.2%)
female_stereotyped	670 (13.4%)	61 (9.6%)	394 (14.3%)	215 (13.5%)
male_stereotyped	759 (15.2%)	93 (14.6%)	453 (16.4%)	213 (13.4%)

Table 3.4: Stereotype distribution per split

3.5 Dataset Extension

I created an addition to the other three datasets to include examples that test a system's ability to detect lexical gender. The dataset extension contains single-sentence examples containing target nouns with male or female lexical gender and examples containing nouns with neutral lexical gender but stereotypical gender associations. To get a mixture of different contexts, I extracted examples from a few smaller corpora accessible through the NLTK corpus module (Bird et al., 2009). These corpora include the Gutenberg, Brown, and Inaugural corpus in the versions that can be accessed through the NLTK module.

The selection of texts taken from Gutenberg includes novels and stories published between the 18th and the 20th century. Literature from this time period tends to stick to typical gender stereotypes, which makes it suitable to measure how well systems

Table 3.5: Extraction rules for examples testing lexical gender classification. **bold**: target noun; blue: noun with neutral lexical gender, red: masculine/feminine lexical gender, gold: referential gender cue.

Example	Extract	Reason
The **cook** tasted the broth.	✓	Target noun tagged `NOUN`
They *cook* every Sunday.	✗	*cook* tagged `VERB`; no target noun
The **dietitian** was also a doctor.	✓	Target noun with neutral lexical gender, no other gender cue (*ambiguous*)
The **widow** was also a doctor.	✓	Target noun with feminine lexical gender (*unambiguous_female*)
The **dietitian** was also a widow.	✗	Gendered word changes the target noun's neutral lexical gender
The **dietitian** revised her plan.	✗	Target noun's neutral lexical gender would be changed by the referential cue (pronoun)
The **widow** lost her keys.	✗	Referential cue (pronoun) provides an extra hint and is therefore not allowed in the dataset

can differentiate between true lexical gender and nouns with neutral lexical gender in a gendered context. The Bible, which is usually included, was taken out due to its length and high use of the word *man*, which would decrease the variety and diversity of the resulting passages. It is furthermore not entirely clear how *man* would be labeled in this specific context since it is often used generically to refer to all of humanity. The Brown corpus includes texts from several genres, including different types of literature, journalism, and administrative texts. Its inclusion helps diversify the dataset. The Inaugural corpus includes 55 inauguration speeches held by previous presidents of the United States. This corpus adds a component to the dataset that includes many instances of lexically gendered plural nouns that are used in a more general context instead of referring to a specific person, such as in "Our journey is not complete until our gay brothers and sisters are treated like anyone else under the law" (Obama, 2013).

The dataset contains 1402 sentences of which each contains one target noun whose linguistic gender should be classified. Every sentence is labeled with the linguistic gender of the target noun and its stereotypical gender association.

This dataset contains multiple varieties of English, including different varieties of written American English, as found in the Brown Corpus, written American and British English from the 18th, 19th, and 20th century in both prose and lyrical form, as found in the Gutenberg Corpus, and spoken American English used specifically in inaugural speeches of US-American presidents. The speakers can be defined as American journalists, politicians, especially presidents, and British and American authors of 18th to 20th century literature.

The dataset was automatically annotated, using lists from Bolukbasi et al. (2016), Garg et al. (2017) and Monsen (2019) . It was then manually checked and post-edited by me, a 25-year-old master's student of linguistics. I am an L1 speaker of German and an L2 speaker of English.

To extract the examples, I used lists to make sure that every sentence contained a noun whose lexical gender can only be identified based on its lexical gender. Lists from Bolukbasi et al. (2016) and Garg et al. (2017) were used to find nouns with neutral lexical gender that were either stereotypically associated with female or male gender or neither. Another list of gendered words from Bolukbasi et al. (2016) and Monsen (2019) were used to identify words carrying lexical gender. To avoid misunderstandings and irrelevant extractions due to polysemy, I went through the lists to eliminate any words that could potentially also refer to an entity that is not a person. These lists combined with a collection of other gender cues, which are described in Tables 4.1 and 4.3), were also used to automatically eliminate any gender cues besides the target noun itself in sentences with neutral target nouns. Additionally, only sentences where the target word was pos-tagged as `NOUN` were extracted. For the full lists, view Appendix A.2.2. Table 3.5 examples of which sentences were and were not included in this part of the dataset.

The dataset extension finds its relevance in testing how systems can detect a noun's gender based on its lexical gender, which is not included in the other datasets

3.6 Complete Dataset

seed	s1	s2	s3	stereotype	dataset	label
dietitian	—	Lou is now a **dietitian** [...]. That's why he is often consulted [...].	—	female	FairTranslate	male
investors	[...] [e]ntrepreneurs sought funding to fuel growth.	Key **investors** identified high-potential start-ups [...].	The strategic involvement of these women was often a catalyst for change [...].	—	Glitter	female
mother	—	My own **mother**'s family had a banshee [...].	—	—	Lexical	female
lawyer	—	No opportunity is given to [...] prisoners to choose their own **lawyer**.	—	—	GeNTE	ambiguous

Table 3.6: Example instances taken from the unified development set; Examples and label names were shortened to save space.

The motivation behind combining the explained datasets was to create a dataset that could be used to evaluate systems on their ability to perform linguistic gender detection. Together, the datasets cover pronouns, titles, lexical descriptions, and lexical gender as possible gender cues.

The columns of the final dataset contain the target noun, the full text passage,

stereotypical gender associations connected to the target noun, the gold label, and the name of the source dataset. Table 3.6 specifies the structure of the final combined dataset.

More specific information on language varieties, text characterisics, and speaker/annotator demographics included in the dataset can be found in the previous subsections that describe its separate components.

In the final dataset, some examples were filtered out due to errors or not fitting the task.

Chapter 4

Methodology

This chapter explains the methods that were used to investigate how the linguistic gender of nouns referring to people can be detected from information given in the surrounding text passage.

To briefly repeat the goal that the methods introduced in this chapter aim to achieve, this thesis investigates the identification of linguistic gender based on explicit gender cues, such as pronouns, lexical gender, or other lexical descriptions. Implicit gender stereotypes can lead to harmful gender assumptions and errors, especially in anti-stereotypical contexts (Gnadt et al., 2025; Savoldi et al., 2025). Systems detecting gender from explicit cues are therefore compared to systems with access to other contextual information, such as attributes, actions, or occupation titles, that might result in decisions made based on gender stereotypes.

This chapter outlines the exact task, describes which cues the systems use and how they are detected, gives a summary of the datasets used, and explains the different systems applied in this thesis.

4.1 Task Overview and Datasets

This section describes the general task this thesis deals with, explains what exactly is meant by detecting linguistic gender and explicit gender cues, and how a system's ability to detect linguistic gender is going to be evaluated.

4.1.1 Task Definition

All systems presented in this section deal with detecting the linguistic gender of nouns referring to a person in English text. The task is to detect the gender that can be deduced from the surrounding text passage, rather than the correct real-world gender of the person behind a noun. If the gender of a noun is not indicated or can not clearly be deducedc from a noun's linguistic environment, it should be classified as `ambiguous`.

This task distinguishes between ambiguous and unambiguous instances. If the gender of a noun is indicated within a given text passage, it should be labeled as `unambiguous_female`, `unambiguous_male`, or `unambiguous_all`.
While `unambiguous_female` and `unambiguous_male` can refer to singular or plural entities, distinguishing between male and female gender, `unambiguous_all` specifically describes plural nouns referring to mixed-gender groups and should be assigned if a text makes it clear that a certain group is made up of at least two different genders.

The most widely-used way of expressing a non-binary gender identity through language is to use gender-neutral language (Rose et al., 2023), such as the *singular they*, which is also used in cases where someone's gender is unknown, to avoid binary gender cues and labels. Therefore, non-binary gender often does not form its own linguistic category, except for cases in which neo-pronouns are used or non-binary gender is explicitly discussed. In this thesis, it is therefore labeled as `ambiguous`, which is valid for any case in which gender-neutral language, i.e. language that avoids assigning binary gender to an entity, would be appropriate.

Regarding all gender categories used in this thesis, it is important to emphasise that it deals with detecting gender from how it is linguistically expressed. While in language, different categories are used to express gender, social and biological gender should be viewed as a spectrum (Bing and Bergvall, 1996; van Anders et al., 2017). Therefore, linguistic gender does not necessarily reflect a person's real-world gender, as it often fails to capture its complexity (Bucholtz, 1999).

4.1.2 Explicit Cues: Components

The systems detecting linguistic gender from explicit cues, which are described in Sections 4.2.1, and 4.2.2, only have access to information that explicitly indicates gender, which can be found in singular pronouns, lexical descriptions, titles, and a noun's lexical gender. Table 4.1 provides an overview of the mentioned types of gender cues. Each of these cues and the methods used to extract them will be described more closely in the following subsections.

Cue Type	**Description**	**Examples: fem**	**Examples: masc**
Pronouns	third-person singular gendered pronouns	she, her, herself	he, him, his, himself
Lexical gender	gender inherent to a word	woman, queen, mother	man, king, father
Lexical cues	words describing a noun's gender	the female journalist the doctor, who was also a mother	the male journalist the doctor, who was also a father
Titles and names (optional)	titles that indicate gender	Ms, Mrs, Miss, Lady	Mr, Sir, Lord

Table 4.1: Types of Cues

Coreference Resolution: Pronouns and Gendered Nouns

Coreference resolution generally describes the task of detecting which references in a text refer to the same entity (Pradhan et al., 2011). These references can come in the form of pronouns or gendered referents that can help classify a noun's linguistic gender. In this thesis, coreference resolution is therefore applied to identify gendered pronouns and/or gendered nouns referring to the target noun. The Maverick coreference resolution system (Martinelli et al., 2024) was chosen due to its state-of-the-art

performance and high efficiency, which results in a system that is easier to run with lower computational resources.

Type	**Example**	**OntoNotes**	**PreCo**
Generic Mentions	*[dogs] can bark*	Only with pronoun/determinate NP, or in headlines	All generic NPs/modifiers annotated
Appositives	*[[Abe], [the chef]]*	Annotated, not coreference	3 mentions (both NPs + larger span)
Copular Predicates	*[he] is [the teacher]*	Not annotated	Referent and attribute are coreferrents
Compound Modifiers	*[Taiwan] authorities*	Only non-adj. proper nouns, excl. nationality acronyms	All annotatable

Table 4.2: Differences in how coreference is defined in OntoNotes and PreCo (summarised table from Porada et al. (2024).

The chosen coreference system was trained on the PreCo dataset (Chen et al., 2018), which fits the task due to its annotation guidelines which generally count appositves and copulative predicates as coreferents (Porada et al., 2024). As shown in Table 4.3, these structures can be relevant to determine a noun's linguistic gender. The guidelines furthermore tend to be more permissive towards nested mentions and compound modifiers, which makes the dataset especially useful for this task compared to others. Since referential gender information can often be assigned in the form of attributes or roles, which are not defined as coreferents for datasets with stricter guidelines (Porada et al., 2024), a system detecting referential gender can profit from these more permissive guidelines.

To extract a noun's referential gender from a coreference cluster, gendered pronouns that represent the head of a reference within the cluster are counted. If pronouns referring to different genders are present, a majority count is applied for singular nouns while both genders are recorded for plural nouns. A similar logic is applied for gendered nouns within the coreference cluster, which will be explained further in Section 4.2.1.

Lists and WordNet Glosses: Lexical Gender

To determine a noun's lexical gender, the system uses a mixture of lists and a system introduced by Bartl and Leavy (2022), which counts gender cues in a word's definition to find its lexical gender.

First, three lists containing key words of neutral, feminine, or masculine lexical gender, taken from Bartl and Leavy (2022), are checked for the target noun. If the noun can be found in one of the lists, the correct lexical gender can immediately be assigned. Otherwise, the first five words of the first two WordNet definitions are searched for gendered words, after which a majority vote decides whether a noun is assigned feminine, masculine, or neutral lexical gender. Figure 4.1 provides a more detailed description of how lexical gender is deduced from WordNet glosses.

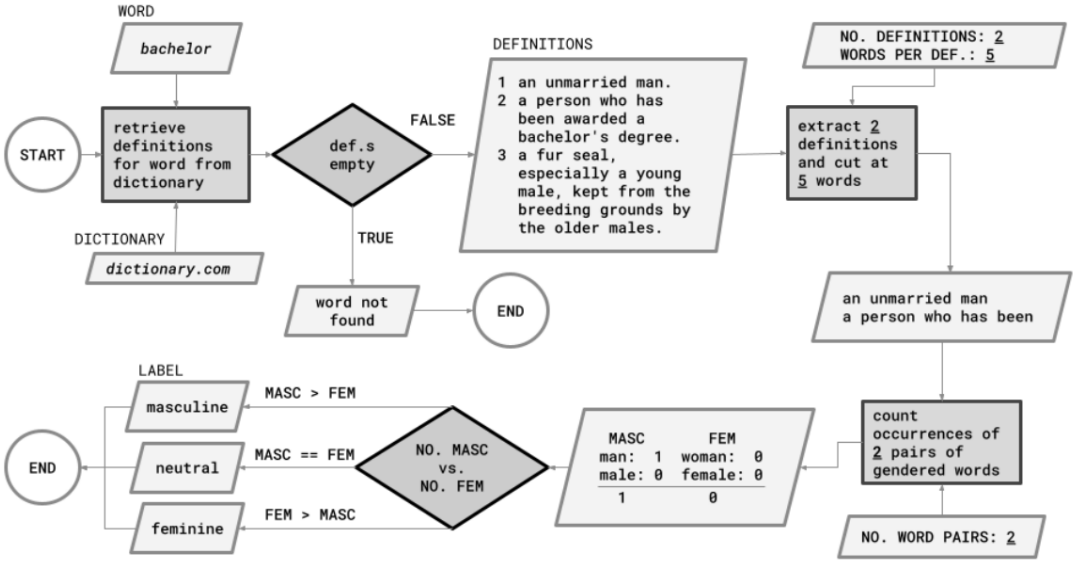


Figure 4.1: Reproduction of Figure 1 from Bartl and Leavy (2022), showing a simplified flowchart of their lexical gender detection algorithm

Dependency Relations: Lexical Cues

For any referential gender cue that cannot be caught by the coreference system, dependency relations connecting the noun and the cue are defined. For a summary of the most important dependency relations, view Table 4.3. Although the PreCo dataset, which the coreference model was trained on, counts appositions and attributes in copulative structures as coreferents (Porada et al., 2024), they are often not identified as such by the coreference system. Furthermore, for plural entities, especially for groups whose members are individually mentioned, it is useful to define how they are syntactically related since, although they belong to the same entity, the components' gender labels can vary. For groups where only some of the members are mentioned, dependency structures become vital as well, since they cannot be identified as the same entity. To see the complete set of dependency rules, please refer to Appendix A.

Titles and Names

Gendered titles, such as *Mr* for men or *Ms* for women, are interpreted as explicit cues for male or female gender. They are the primary cue that is taken into account regarding names referring to nouns.

As an optional feature, first names can be taken as cues too. If this option is activated, gender guesser (Saeta Pérez, 2016), a lookup tool that works with a database based on gender-c (Michael, 2007), is used to identify the gender that is usually associated with a name. This tool categorises names as either `female`, `male`, `mostly_female`, `mostly_male`, `andy` (androgynous), or `unknown`. If a name is labeled as `female` or `male`, it is counted as a cue for the respective gender. The other categories, `mostly_female`, `mostly_male`, `andy`, and `unknown`, are not taken as a signal for any linguistic gender.

However, since linguistic gender, including pronouns, does not always intersect with the gender a name is typically associated with, names can only be interpreted as explicit cues to some degree and are therefore not taken into account for the main evaluation.

Dependency Structure	Dependency Relation	Examples: fem	Examples: masc
Modifiers	amod ← noun compound ← noun	the female journalist the woman journalist	the male journalist the man journalist
Copulative structures	nsubj ← cop verb → attr nsubj ← cop verb → acomp	The doctor was a woman. The doctors were all female.	The doctor was a man. The doctors were all male.
Relative clauses	relcl → cop verb → attr relcl → cop verb → acomp	the dancer, who was a woman the dancers, who were all female	the dancer, who was a man the dancers, who were all male
Appositions	noun → appos	The teachers, all women, were tired	The teachers, all men, were tired
Group-membership verbs	nsubj ← verb → dobj nsubj ← verb → prep → pobj	The team includes several women. The team consists of women.	The team includes several men. The team consists of men.

Table 4.3: Main Dependency Relations for Lexical Cues (Draft)

4.1.3 Datasets

This section provides a short overview of the datasets used in this thesis. The full dataset used to test (40%), train (50%) and develop (10%) the different systems consists of three datasets from the field of gender-fair machine translation which cover different forms of referential gender and a self-created dataset extension that covers lexical gender. For more detailed information on the datasets, refer to Chapter 3. For the trained classifiers, the training and development set are combined to form a larger training set. This split is due to the fact that I use k-fold cross validation for those classifiers, which eliminates the need for an extra development set.

Glitter

The Glitter dataset (Pranav et al., 2025) was created for gender-fair machine translation from English to German. It contains 909 unique passages, containing 147 different gender-ambiguous seed nouns, meaning that the seed nouns themselves do not contain lexical gender. Each instance consists of four sentences, one of which contains the seed noun in its plural form, while the other three consist of two preceding sentences and one trailing sentence. All four sentences can, but do not always, contain cues for the seed noun's gender. All instances are annotated with the gender of the seed noun, which can either be `ambiguous`, `unambiguous_female`, `unambiguous_male`, or `unambiguous_all`. It contains text taken from Wikipedia and synthetic passages.

In this thesis, the dataset is used to test a system's ability of recognising referential gender from lexical descriptions and context. Since the seed nouns are all plural, the

sentences do not contain gendered pronouns that could point towards a specific gender. Furthermore, because the seed nouns do not contain lexical gender, their gender can only be detected through words referring to the seed noun.

FairTranslate

FairTranslate (Jourdan et al., 2025), a dataset created in the context of gender-fair machine translation from English to French, contains 2418 sentence pairs, based on 62 singular seed nouns, of which some include male, female, or inclusive pronouns (singular they) while others do not. The set includes information on the gender of the seed nouns, classifying them as either male, female, or ambiguous and providing information on whether these nouns fit male or female gender stereotypes. Since the seed nouns, which are the nouns that should be classified, were only provided in their French version, the English nouns, which are relevant to this task, had to be extracted 3. The dataset contains shorter examples that consist of one sentence and longer examples which contain multiple sentences. It consists entirely of synthetic data.

FairTranslate will be used to cover the systems' ability to detect the gender of singular nouns based on the pronouns used in connection to them since, in this dataset, gender is largely determined through pronouns.

GeNTE

GeNTE (Piergentili et al., 2023) is a dataset for gender-fair machine translation from English to Italian. It consists of 1500 sentence pairs, containing ambiguous, male, and female examples. The sentences were extracted from the EuroParl corpus (Koehn, 2005) and contain several kinds of potential gender cues, including gendered titles, pronouns, and other lexical descriptions of gender.

GeNTE is therefore the dataset that best covers the different capabilities on which the systems should be tested. It is, however, especially relevant for testing the systems' ability to deduce gender from titles, which is not covered by the other datasets.

Dataset Extension

The dataset extension, which I created to test the systems' ability to use lexical gender as a cue, consists of 1402 sentences with nouns that contain either male or female lexical gender, nouns that have neutral lexical gender but are often stereotypically associated with a gender, and nouns with neutral lexical gender that are not commonly associated with a specific gender. The passages do not contain any explicit linguistic gender cues besides the noun's lexical gender. The sentences were extracted, unedited, from three corpora, which are Gutenberg, Brown, and the Inaugural corpus, accessed through the `nltk.corpus`-module. The extension therefore consists of original occurrences rather than minimal pairs.

The dataset extension therefore tests the systems' ability to handle lexical gender and the difference between words that contain lexical gender and those that do not but are perceived as being connected to a specific gender.

4.2 System Architectures

This section details the design and architectures of the different systems applied. To avoid classifications based on gender stereotypes, a rule-based and a feature-based classifier are applied as systems working exclusively with explicit gender cues. As counter-examples, I fine-tuned a BERT model and prompted two generative LLMs, Qwen 2.5 3B and Qwen 2.5 72B (Yang et al., 2024), which are given access to all contextual information. Another system combines the rule-based classifier with a large language model (Qwen 2.5 3B). First, the rule-based system classifies all nouns. For all nouns identified as `ambiguous`, the LLM is provided with the lexical gender cues found in the current text passage. It is then asked to assign them to the correct noun.

4.2.1 Rule-Based Classifier

The rule-based system presented here applies a pipeline including coreference resolution to collect references pointing to the same entity, lists and WordNet glosses to determine a noun's lexical gender, and dependency relations to collect lexical gender cues found in constructions such as modifiers, appositions, or relative clauses. While each component is described more closely in Chapter 4.1.2, the following paragraph aims to give a short overview of the system's workflow.

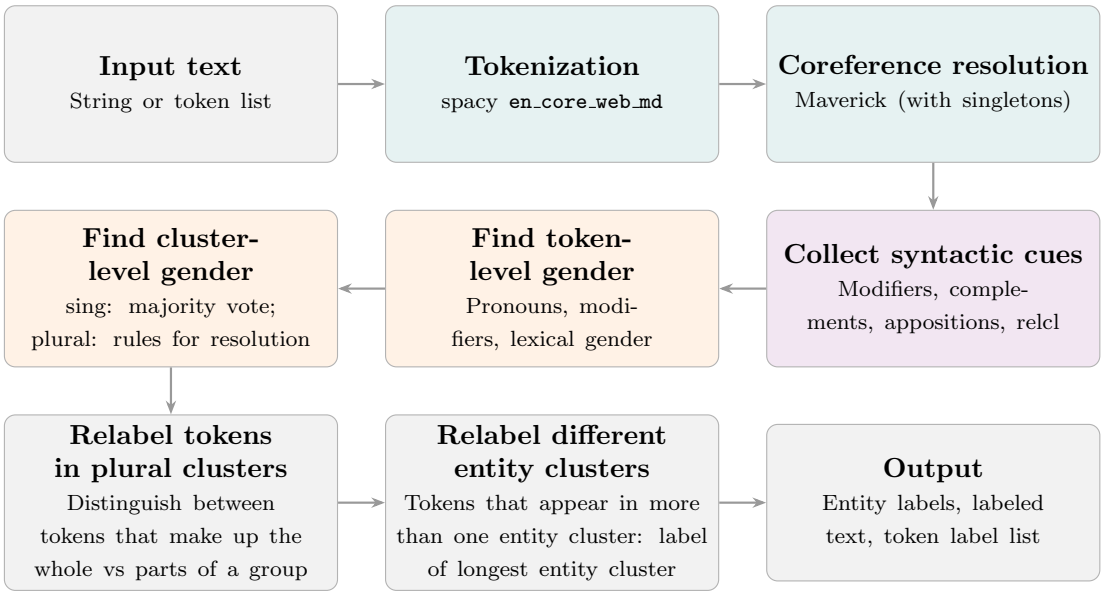


Figure 4.2: Pipeline Overview

To preprocess the input text, it is tokenised and sorted into entity clusters, using coreference resolution, and parsed using spacy `en_core_web_md`, which provides part-of-speech tags, named entity recognition, and morphological features including number (singular/plural), which the system uses to distinguish between individual and group-level gender resolution.

As a second step, the system collects indexes of words that could potentially be a relevant gender cue. To find all names mentioned in the text, all indexes of tokens labeled as `PERSON` by spacy's NER-tagger are added to a list. Next, indexes of gendered adjectival modifiers, such as *female* or *male*, and gendered parts of compounds, such

as *woman* in *woman warrior*, are recorded as well. After that, several syntactic structures (view Table 4.3 for descriptions and examples) that might connect a noun to a gendered complement, such as copula constructions, relative clauses, and appositions, are identified and recorded.

With these steps completed, the system can start the classifying process, which is first done separately for each token and then again for each entity cluster. Since this classifier aims to detect the gender of nouns referring to people, only tokens that are either marked as `PERSON` by the NER component or pos-tagged as noun or proper noun while also having *person* as a hypernym in WordNet are considered. As depicted in Figure 4.4, all tokens that are found through NER are first checked for gendered pronouns and titles that indicate gender, such as *Mr* or *Ms*. If no titles are found, the name itself can optionally be checked through the `gender-guesser` library (Saeta Pérez, 2016). For all other nouns referring to a person, gender is assigned using a priority-based system that uses three different types of cues, which are gendered pronouns, gendered modifiers, and lexical gender. As described in Figure 4.3, if a gendered pronoun is detected, it determines the gender of the target noun. Otherwise, the system takes any gendered modifiers into account to determine the target noun's gender. If no gendered modifiers are present, the target noun's lexical gender determines how the target noun is classified. If no explicit gender cues are found, the token is labeled as *ambiguous*.

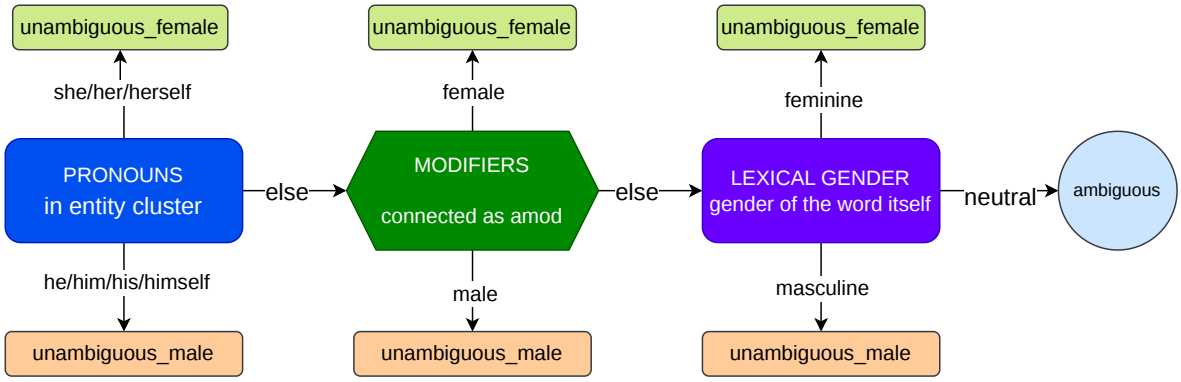


Figure 4.3: Token-Level Gender Classification (Common Nouns)

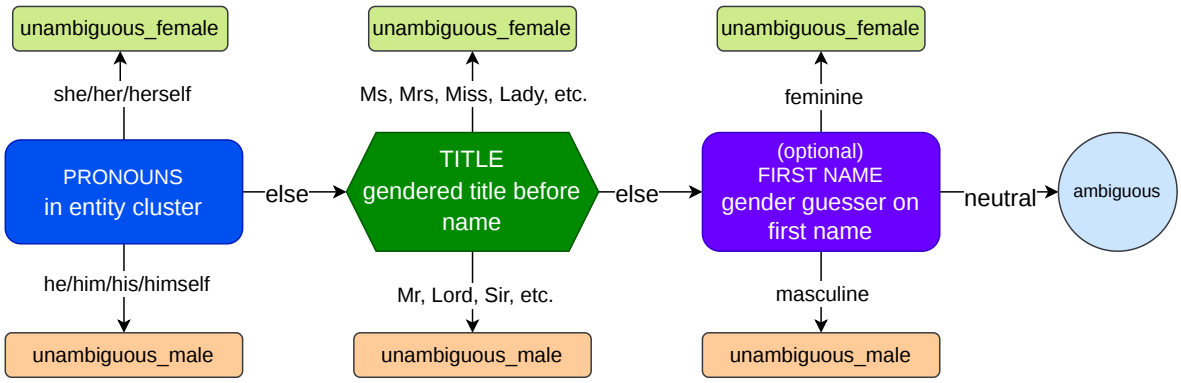


Figure 4.4: Token-Level Gender Classification (NER Person)

Once all tokens have been assigned an initial label, the system moves on to resolv-

ing the gender of each entity cluster. For clusters that only contain references in their singular form, a majority count is applied, which is especially relevant for cases where the single reference labels might contradict each other due to mislabeling. The most common gender label that is not `ambiguous` is applied to all reference heads of the entity cluster. If the majority count results in a tie, the noun is labeled as `ambiguous`. For clusters that contain plural references, all gender labels within the cluster are recorded. If labels for male and female referents can be found within the same cluster, the cluster label generally becomes `unambiguous_all`. After this step, every referent in a plural cluster needs to be evaluated to see whether it represents the whole or only a part of an entity, which determines the final labels of the referents within a plural entity cluster. This is done using syntactic rules defined in Appendix A. Finally, tokens that are part of multiple coreference clusters and have therefore been assigned multiple labels are assigned the label that is the most specific according to a predefined label priority order, resulting in the final labels.

Label order (least to most specific): ambiguous, unambiguous_male, unambiguous_female, unambiguous_all

4.2.2 Feature-Based Classifier (Logistic Regression)

The feature-based classifier represents the second example of classifiers that only have access to explicit gender cues. It uses a logistic regression algorithm due to its interpretability and the simple manner in which the features interact. The features for each token are transformed into one-hot vectors and the classifier is evaluated using 5-fold cross validation. It is trained on 60% of the dataset and evaluated on 40%. The specific features it uses are described below.

Features

- **Lexical Gender**: This feature checks a token's lexical gender in the way that is explained in Section 4.1.2. Depending on the result, its value can either be `masc`, for masculine lexical gender, `fem`, for feminine lexical gender, `neutral`, or `not_found` if the token cannot be found in the lists or WordNet.
- **Pronouns**: This feature indicates the gender that can be deduced from gendered pronouns within the token's entity cluster. As described in Chapter 4.1.2, it collects all gendered pronouns within the cluster and applies a majority count to arrive at a gender label, which can either be `neutral`, `male`, or `female`.
- **Modifiers**: This feature checks whether a noun is connected to a modifier such as *female* or *male*, or part of a compound construction with a gendered word such as *man* or *woman*. The outcome label can be neutral if no modifier is found, or otherwise `female`, `male`, or `both`.
- **Combined: Number, Entity Genders, and Token-Level Gender of the Current Token**: This combined feature indicates which gender categories can be found for the different reference heads within a token's coreference cluster, whether the token itself is singular or plural, whether the other reference heads within the cluster are singular or plural, and the current token's token-level gender (sourced as described in Figure 4.3).

Ambiguous (no cue)

A **nurse** prepares medical files before a routine check-up.

lexical: neutral *modifier:* none *cluster:* sing_noplural_n_neut *pronoun:* neutral *masc path:* none *fem path:* none

Gendered Pronouns

This **journalist** is known for immersive reporting, giving a voice to marginalized populations. He inspires respect and admiration.

lexical: neutral *modifier:* none *cluster:* sing_noplural_m_masc *pronoun:* masc *masc path:* none *fem path:* none

The **butcher** prepared a special order for a customer hosting a barbecue, including tips for cooking skewers. A few days later, the customers returned to thank her for the delicious order.

lexical: neutral *modifier:* none *cluster:* sing_noplural_f_fem *pronoun:* fem *masc path:* none *fem path:* none

Lexical Gender

Does your **husband** have a car??

lexical: masc *modifier:* none *cluster:* sing_noplural_m_masc *pronoun:* neutral *masc path:* none *fem path:* none

Lexical Reference (dependency path)

Like Vargas, each of his brothers are decorated **veterans** [. . .]

lexical: neutral *modifier:* none *cluster:* plural_plural_n_neut *pronoun:* neutral *masc path:* up_dobj_down_nsubjpass_down_prep_down_pobj *fem path:* none

Lexical Reference (coreference)

The beauty industry has seen a remarkable evolution with the advent of social media platforms. Tutorials and reviews influence consumer choices significantly. Beauty **influencers** have dominated the industry, shaping trends and brand perceptions with their engaging content. The impact of these women is not only seen in sales figures but also in how brands tailor their products to meet diverse expectations.

lexical: neutral *modifier:* none *cluster:* plural_plural_f_neut *pronoun:* neutral *masc path:* none *fem path:* none

Gendered Modifier

Regarded as one of Australia's leading contemporary women artists, Minnie ranks alongside other notable Indigenous female **painters** Dorothy Napangardi, Gloria Petyarre and Kathleen Petyarre.

lexical: neutral *modifier:* fem *cluster:* plural_plural_fn_fem *pronoun:* neutral *masc path:* none *fem path:* is_fem

Table 4.4: Examples of sentences in their feature-based representations.

If the feature is `plural_plural_f_neut`, it means that the noun itself is plural and that the other nouns within the cluster are also plural. *neut* signalises the target noun's token-level gender, while *f* symbolises the gender labels of the other referents in the cluster, which, in this case, is *female*. Such a label would therefore indicate female gender through a referential gender cue. A specific example of this can be found in Table 4.4.

- **Dependency Path**: The last two features show the directed dependency path from the current token to the syntactically closest gender cue, which is determined by the number of steps required to get from the token to the cue. While one of

the features does this for the closest female gender cue, the other feature indicates the path to the closest male gender cue.

4.2.3 Fine-Tuned BERT

As an example of a classifier that is trained on the full text instead of only explicit cues, a BERT (Devlin et al., 2019) model (distilbert-base-uncased) (Sanh et al., 2020) was finetuned for classification. Since each text passage in the dataset only contains one noun to classify, that noun was marked by inserting a special token, `[CUE]`, in front of it. The input was then tokenized into subtokens, using the WordPiece (Schuster and Nakajima, 2012) tokenizer, and trained and tested on three seeds (14, 98, 164). To arrive at the final labels, the original tokens were reconstructed from the subtokens, using a majority vote in case of label disagreement between subtokens that originated from the same original token.

4.2.4 LLM as a Second Filter

This option combines the rule-based system with Qwen 2.5-3B. Linguistic gender is sometimes determined by contextual cues that cannot necessarily be found through dependency paths or coreference resolution. In these cases, the LLM (Qwen 2.5 3B) helps establish connections between the target noun and the relevant gender cue.

First, the rule-based system classifies all nouns. If the rule-based system classifies a noun as `unambiguous_female`, `unambiguous_male`, or `unambiguous_all`, the step that involves the LLM is skipped. However, all nouns that the rule-based system classifies as `ambiguous` are passed on to an additional filter where an LLM is prompted to decide whether a given gender cue refers to the target noun, or not. For this, it is provided with all lexical gender cues that can be found within a window of 50 tokens before and after the token that is being classified. For each cue, it is then asked to decide whether the cue influences the linguistic gender the target noun. A possible case of this can be seen in the following example.

Example 1. *The [women's] annual ballet recital was held downtown, and the auditorium was packed with eager families and dance enthusiasts. From the youngest pupil to the most seasoned professional, every performance was met with thunderous applause. The <<<dancers>>> effortlessly executed the complex choreography with poise and grace. Their instructor, a veteran of many dance stages, watched with pride and satisfaction from the wings.*

In this sentence, taken from the Glitter dataset (Pranav et al., 2025), the word *women* signalises that the word *dancers* should be classified as female. However, the two words are in different sentences, gender is implied instead of explicitly described, and the gender cue and the target noun are not coreferents. The rule-based system would therefore fail at identifying the linguistic gender of *dancers* and label it as `ambiguous`.

The LLM is then provided with the sentence combined with the following prompt for each of the cues:

```
In this specific text passage, does the gender cue [women]
influence the gender of the word <<<dancers>>>? Reply with one
word.
```

If the LLM's answer contains the word *yes*, the gender of the explicit cue is recorded. When the LLM has been asked about all cues in connection to the relevant token, the following rules determine the final gender label the token will be assigned. If a cue for female gender and a cue for male gender are assigned to the same token, it is classified as `unambiguous_all`. If no cue is assigned to the token, it is classified as `ambiguous`. If one or more cues of the same gender are assigned to the token, that gender gets assigned to the token, except for when one or more of the cues are preceded by a verb or quantifier that implies that the cue only refers to a part of a group, in which case the token is classified as `unambiguous_all`. In this context, explicit cues can be nouns containing lexical gender or adjectives signifying gender. A list of possible cues can be found in Appendix A.2. The full prompt is in Appendix B.2.

4.2.5 Baselines

LLM (Qwen 2.5-3B and Qwen 2.5-72B)

To test how all systems perform, Qwen 2.5 72B (Yang et al., 2024), the language model that performed best for the initial annotation of Glitter (Pranav et al., 2025) (weighted F1 of 78%), including the prompt and settings used by Pranav et al. (2025), is taken as a baseline. To compare how larger and smaller large language models handle the task, Qwen 2.5-3B was included as an example of a smaller LLM. For both models, the temperature is set to 0, meaning that the LLM always picks the most probable output. The original prompt is written in a few-shot format and edited to include examples from each dataset and for each type of gender cue. The opening of the prompt is formulated as shown in the example below, which includes one of the examples provided to the LLM. The full edited prompt can be found in Appendix **??**.

```
You are a data filtering assistant. Your task is to determine
whether the gender of the seed word is disambiguated by the
context. Avoid making assumptions based on names or roles.
Include the words "AMBIGUOUS" or "UNAMBIGUOUS" in your response.
Keep your response concise and to the point.

CLEARLY DISAMBIGUATED EXAMPLES:

Female Painters Example:
Seed: "painters"
Context: "The annual town arts festival was in full swing,
attracting artists from across the region. The main square was
filled with colorful tents showcasing a variety of artistic
talents. Two young painters set up their easels side by side,
capturing the vibrant scene on their canvases. Visitors stopped
to admire the work of these young women, mesmerized by the
vibrant hues and bold strokes."
Output: The context explicitly refers to the painters as "young
women". UNAMBIGUOUS (FEMALE).
```

Majority Baseline

The majority baseline guesses `ambiguous` for every token since `ambiguous` represents the majority class in all datasets. Furthermore, the classifiers based on explicit cues have a tendency to guess `ambiguous` since that is the default answer if no explicit cues are found. This baseline therefore helps evaluate their performance.

Random Baseline

A random baseline, which always picks a random label out of all labels used in the relevant part of the dataset, is added for further comparison and context on how well the systems perform.

4.3 Evaluation Metrics

The main evaluation and comparison will be based on the weighted f_1-score since it was the metric used by Glitter (Pranav et al., 2025), which inspired this thesis, to compare different LLMs used to annotate the dataset with gender labels. The datasets, besides Glitter, which contains a very small amount of `unambiguous_all`-instances, do not show any heavy imbalances, which makes the metric appropriate for evalutation, especially since every instance of the dataset is relevant for gender-labeling. Outside-labels, which often cause an imbalance in the label distribution, do therefore not exist. For the exact label distributions, view Tables 3.1 and 3.3.

Chapter 5

Results

This chapter gives an overview of the most important results. It begins by comparing the different systems on the full dataset, after which it inspects the results for each dataset component separately. Finally, it contrasts performance on stereotypical and anti-stereotypical examples, i.e. examples in which the target noun's gender does or does not match the stereotype associated with the noun.

The complete results, including precision, recall, f_1, and accuracy scores for all systems, datasets, and categories can be found in Appendix **??**. More detailed results and observations on the systems' performances regarding types of gender cues and stereotypes are described in Chapter 6.

The test set is evaluated by text passage. Each text passage contains a target noun, whose linguistic gender needs to be labeled.

Disclaimer: The systems applied in this thesis differ in the degree to which they had access to the training or development split of the dataset. While the rule-based classifier mostly relies on out-of-the-box tools and the LLM decoders are only provided with a few examples from the development split, the feature-based classifier and the DistilBERT classifier were trained on the training and development split of the dataset. Since DistilBERT was fine-tuned with access to the full text, it found short cuts in the data that are not transferrable to unseen data, which means that its strong performance on many components cannot be taken at face value. These shortcuts are discussed in more detail in Chapter 6.

5.1 Overall Performance

Baselines				
	Majority	**Random**	**Qwen-3B**	**Qwen-72B**
Weighted F_1	0.237	0.298	0.533	0.879
Systems				
	Rule	**LogReg**	**DistilBERT**	**Rule+LLM**
Weighted F_1	0.837	0.835	**0.927**	0.845

Table 5.1: Weighted F_1 on the Full Test Set

The following section discusses the results achieved by all classifiers on the entire test set, which is made up of Glitter (Pranav et al., 2025), FairTranslate (Jourdan et al., 2025), GeNTE (Piergentili et al., 2023), and an extension of the dataset that I added to test the systems' ability to detect linguistic gender based on lexical gender. A more detailed description of the test set and the different datasets it consists of can be found in Chapter 3.

While all systems presented in this thesis show better overall performance than three of the baselines, which are the two simple baselines (majority and random) and the smaller LLM (Qwen-3B Instruct), only the DistilBERT classifier manages to outperform the larger LLM (Qwen-72B Instruct), which was originally used to pre-annotate the Glitter (Pranav et al., 2025) dataset.

Overall, the DistilBERT classifier achieves the highest scores across all categories, with a weighted F_1 of 0.923, while the three systems based on explicit gender cues (rule-, and feature-based classifiers, plus the combination of the rule-based system and Qwen 3B-Instruct) demostrate relatively similar performances, with weighted F_1-scores ranging from 0.835 to 0.845.

While the larger LLM is able to compete with all other classifiers, with a weighted f_1 of 0.879, the smaller LLM demonstrates results that are significantly lower, with a weighted f_1 of 0.533.

5.2 Performance by Datasets and Different Types of Gender Cues

As mentioned in Chapter 3, FairTranslate uses only pronouns to indicate linguistic gender while in the Glitter dataset, gender can only be deducted from lexical descriptions or contextual implications, meaning that the seed nouns do not have lexical gender and that their gender can also not be deducted from gendered pronouns. The lexical-gender extension tests a system's ability to work with nouns that have lexical gender while GeNTE presents a mixture of different gender cues, including pronouns, titles, lexical descriptions, and to a very low degree, lexical gender.

For an overview of the results the different systems and baselines achieved on the different datasets, view Figure 5.1.

The results for the different components of the test set show that while all systems demonstrate relatively high performance on passages where gender is indicated through pronouns (FairTranslate), they seem to have difficulties with passages where gender is only given through lexical descriptions and context (Glitter). While the DistilBERT classifier performed best for most datasets, it was outperformed by the rule-based system for the dataset extension that looks at lexical gender. However, the difference between both scores is only minimal, falling into the confidence interval of the results achieved by DistilBERT, and is therefore not significant.

While gender indicated through pronouns and lexical gender was handled well by all systems, DistilBERT and Qwen 2.5-72B seemed to have a clear advantage for gender indicated through lexical references. DistilBERT showed the best performance for all datasets besides the lexical gender extension, where DistilBERT was outperformed by the rule-based classifier. Exact numbers and rankings will be discussed in following subsections.

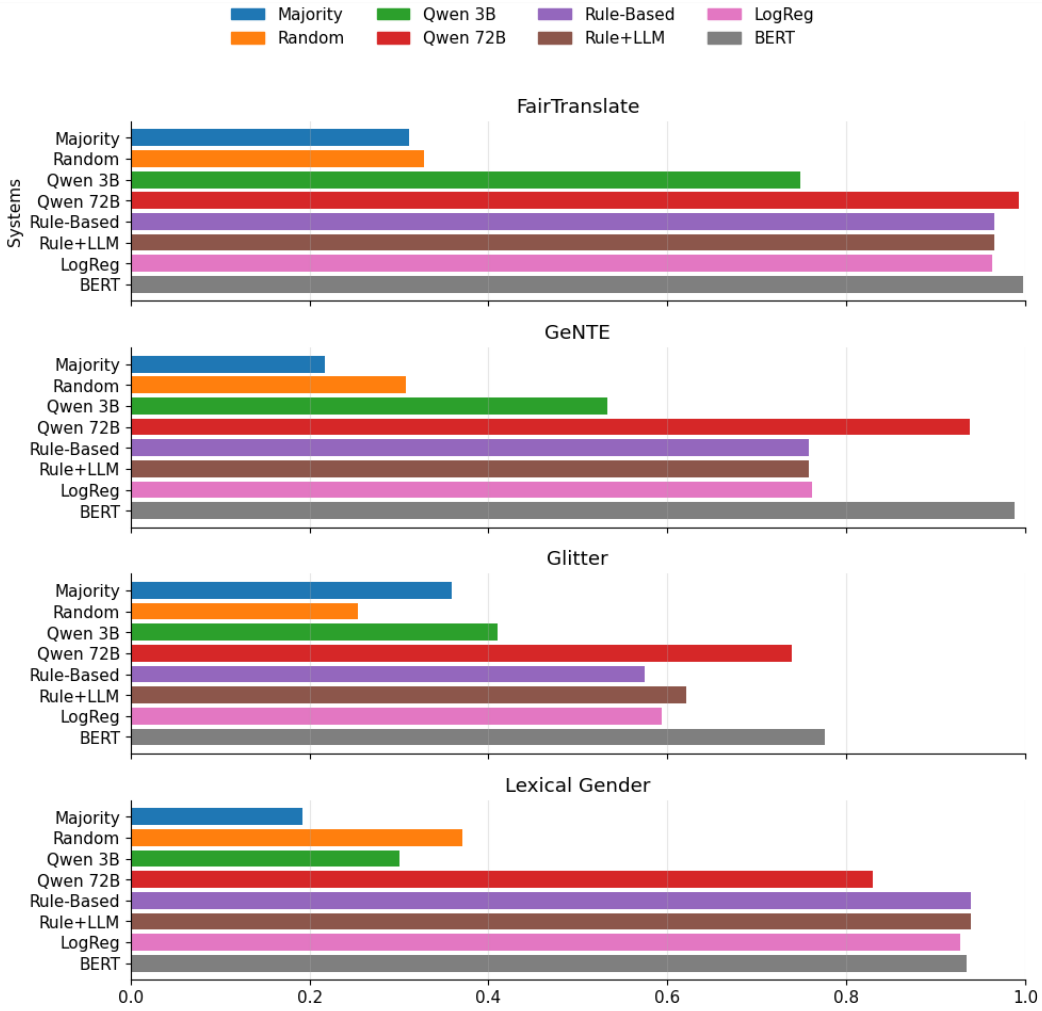


Figure 5.1: Weighted F_1 for Different Datasets and Systems

5.2.1 FairTranslate

In FairTranslate, gender is indicated through pronouns. For a more detailed description of this dataset, refer to Chapters 3.2 and 4.1.3.

All systems presented in this thesis show high results for FairTranslate, with f_1-scores above 0.95 for all categories and systems.

Overall, as can be seen in Table 5.2, DistilBERT achieves the highest results, with a weighted F_1 of 0.998. Concerning the systems based on explicit cues, the rule-based system performs slightly better than the feature-based classifier. All systems show relatively balanced results across the different categories. Since the LLM extension to the rule-based classifier only concerns lexical gender cues, it cannot be expected to improve the results on this pronoun-focused dataset.

Therefore, detecting given gender evidence in the form of pronouns seems to be doable for systems working only with explicit cues and systems with access to the full text. It is, however, also important to state that FairTranslate only consists of sentences where pronoun assignment is not ambiguous while the sentence structures and pronoun positions are repetitive, which makes classification easier.

System	ambiguous ($n = 259$)	unamb. female ($n = 139$)	unamb. male ($n = 142$)	weighted avg ($n = 540$)
Majority	0.648	0.000	0.000	0.311
Random	0.379	0.276	0.288	0.328
Qwen-3B	0.798	0.713	0.693	0.749
Qwen-72B	0.992	0.993	0.993	0.993
DistilBERT	**0.998**	**0.996**	**1.000**	**0.998**
Rule	0.964	0.963	0.968	0.965
LogReg	0.962	0.959	0.968	0.963
Rule+LLM	0.964	0.963	0.968	0.965

Table 5.2: FairTranslate F1 scores by category

It can therefore be concluded that the coreference component of the rule- and feature-based systems succeeds at identifying linguistic gender from pronouns. DistilBERT's high results show that a classifier can be trained to detect linguistic gender from pronouns without using coreference resolution as an intermediate step. However, for more ambiguous cases, the coreference-based system shows clear advantages, which are discussed in Chapter 6.

5.2.2 GeNTE

In GeNTE, linguistic gender is indicated through a mix of different explicit cues, including titles, pronouns, and lexical references. Every example only contains cues that point towards the same gender. Therefore, an example can either contain only female cues, only male cues, or no cues at all. For more detailed information on the dataset, refer to Chapters 4.1.3 and 3.3.

While the DistilBERT classifier and Qwen-2.5 72B show very high scores for GeNTE, with F_1-scores of 0.988 and 0.938 respectively, the systems that only use explicit cues perform significantly worse, with scores between 0.758 and 0.762. The reasons behind these lower scores are further in Chapter 6.

System	ambiguous ($n = 60$)	unamb. female ($n = 94$)	unamb. male ($n = 88$)	weighted avg ($n = 242$)
Majority	0.000	0.560	0.000	0.217
Random	0.252	0.348	0.303	0.308
Qwen-3B	0.563	0.731	0.300	0.533
Qwen-72B	0.878	0.995	0.920	0.938
DistilBERT	**0.974**	**1.000**	**0.983**	**0.988**
Rule	0.563	0.874	0.766	0.758
LogReg	0.573	0.867	0.777	0.762
Rule+LLM	0.563	0.874	0.766	0.758

Table 5.3: GeNTE F1 scores by category

While the classifiers based on explicit cues perform well for `unambiguous_female`, they score much lower for `ambiguous` and `unambiguous_male`. This imbalance can to

some degree be traced back to this dataset containing male pronouns that are interpreted as the generic masculinum in the English source text and therefore labeled as ambiguous (Piergentili et al., 2023). However,the systems working with explicit cues cannot differentiate between a generic masculine pronoun and one that indicates male gender. This will, however, be discussed in more detail in Chapter 6.

Although this dataset contains lexical gender cues, the LLM extension to the rule-based classifier does not make a difference to the results. This is likely connected to the fact that this dataset often indicates gender through a combination of multiple cues. If the rule-based classifier detects a different cue, such as a gendered pronoun, before it looks for lexical gender cues, the extension is never activated. For more detailed information on when the LLM extension is activated, refer to Chapter 4.2.4.

The results might indicate that DistilBERT and Qwen 2.5-72B are able to distinguish generic masculine pronouns from masculine pronouns that indicate gender while the rule- and feature-based classifiers cannot make this distinction. This assumption will be investigated more thoroughly in Chapter 6.

Since this dataset contains multiple cue types, conclusions on how different cue types affect the results can be found in Chapter 6 where the scores are inspected per cue type.

5.2.3 Glitter

In Glitter, gender is indicated through lexical references. All target nouns are plural. For a more detailed picture of the dataset, refer to Chapters 4.1.3 and 3.4.

Glitter represents the dataset on which all tested systems, not including the baselines, score the lowest while also showing a large gap between the performance of classifiers with access to the full context, i.e. DistilBERT and Qwen 2.5-72B, and the classifiers based on explicit cues. While DistilBERT achieves the highest f_1 of 0.776, the rule-based system only achieves an f_1 of 0.575 while the feature-based classifier achieves a slightly higher score, with a score of 0.594, and the rule-based system with the LLM extension represents the classifier with the highest f_1-score among the systems that only focus on explicit cues, with a score of 0.622.

System	ambiguous ($n = 205$)	unamb. all ($n = 32$)	unamb. female ($n = 79$)	unamb. male ($n = 76$)	weighted avg ($n = 392$)
Majority	0.687	0.000	0.000	0.000	0.359
Random	0.309	0.089	0.235	0.198	0.254
Qwen-3B	0.668	0.100	0.238	0.024	0.410
Qwen-72B	0.789	0.456	**0.805**	0.655	0.739
DistilBERT	**0.829**	**0.578**	0.769	**0.764**	**0.784**
Rule	0.732	0.000	0.530	0.440	0.575
LogReg	0.746	0.000	0.542	0.488	0.594
Rule+LLM	0.752	0.000	0.583	0.572	0.622

Table 5.4: Glitter F1 scores by category

Since lexical gender and pronouns fall away as identifiers of the target nouns' linguistic gender, the systems using explicit cues must rely on dependency relations and words identified as coreferents. While the dependency relations only work for specific

cases, as explained in Section 4.3, the coreference system often does not identify a noun's gendered referent as the same entity. Even if these components were more stable, some examples only contain contextual cues that are not coreferents and cannot be identified through pre-defined dependency paths. This dataset is therefore especially difficult for classifiers that only have access to explicit linguistic gender cues.

Another aspect worth pointing out is that all classifiers based on explicit cues fail to correctly identify mixed-gender groups (`unambiguous_all`). Reasons for this will be discussed in Chapter 6.

Overall, the results indicate that systems trained on the full context have a clear advantage over systems based on explicit cues for cases where gender is indicated through lexical references.

5.2.4 Lexical Gender Extension

This part of the dataset contains gendered target nouns, such as *queen* or *king* and gender-neutral target nouns that are stereotypically associated with male or female gender. It therefore tests the systems' ability to identify a noun's lexical gender and to distinguish between a gendered noun and stereotypical associations. More detailed information can be found in Chapters 4.1.3 and 3.4.

The lexical-gender extension of the dataset represents the only part of the test set where a system based on explicit cues, namely the rule-based classifier, with an f_1 of 0.939, performed better than DistilBERT, with an f_1 of 0.935. the mean here is actually 0.935). However, as mentioned in the introduction of this section, the difference between both f_1-scores is so minimal that both lie within the 0.95 confidence interval of the DistilBERT classifier, which was fine-tuned on three different seeds.

System	ambiguous ($n = 127$)	unamb. female ($n = 152$)	unamb. male ($n = 142$)	weighted avg ($n = 421$)
Majority	0.000	0.531	0.000	0.192
Random	0.341	0.383	0.384	0.371
Qwen-3B	0.526	0.218	0.188	0.301
Qwen-72B	0.784	0.904	0.793	0.830
DistilBERT	0.918	0.9951	0.933	0.935
Rule	**0.910**	**0.987**	**0.913**	**0.939**
LogReg	0.897	0.983	0.892	0.927
Rule+LLM	**0.910**	**0.987**	**0.913**	**0.939**

Table 5.5: Lexical Gender Extension F1 scores by category

Compared to the other datasets, this part represents the only part where the larger Qwen-model's f_1-score (0.830) is more than a decimal lower than DistilBERT's (0.935) and the rule- and feature based classifier's f_1 (0.937).

The results indicate that a mixture of the system introduced by Bartl and Leavy (2022) and pre-defined lists of nouns with lexical gender suffice to distinguish lexical gender from stereotypical gender associations. Although the larger decoder shows high F1 scores, the results show that the reasoning abilities of a smaller decoder, i.e. Qwen 2.5-3B, are not enough to distinguish lexical gender from stereotypes.

Overall, lexical gender can be successfully identified by systems that do not have access to the full context.

5.3 Sterotypical vs Anti-Stereotypical Gender Assignment

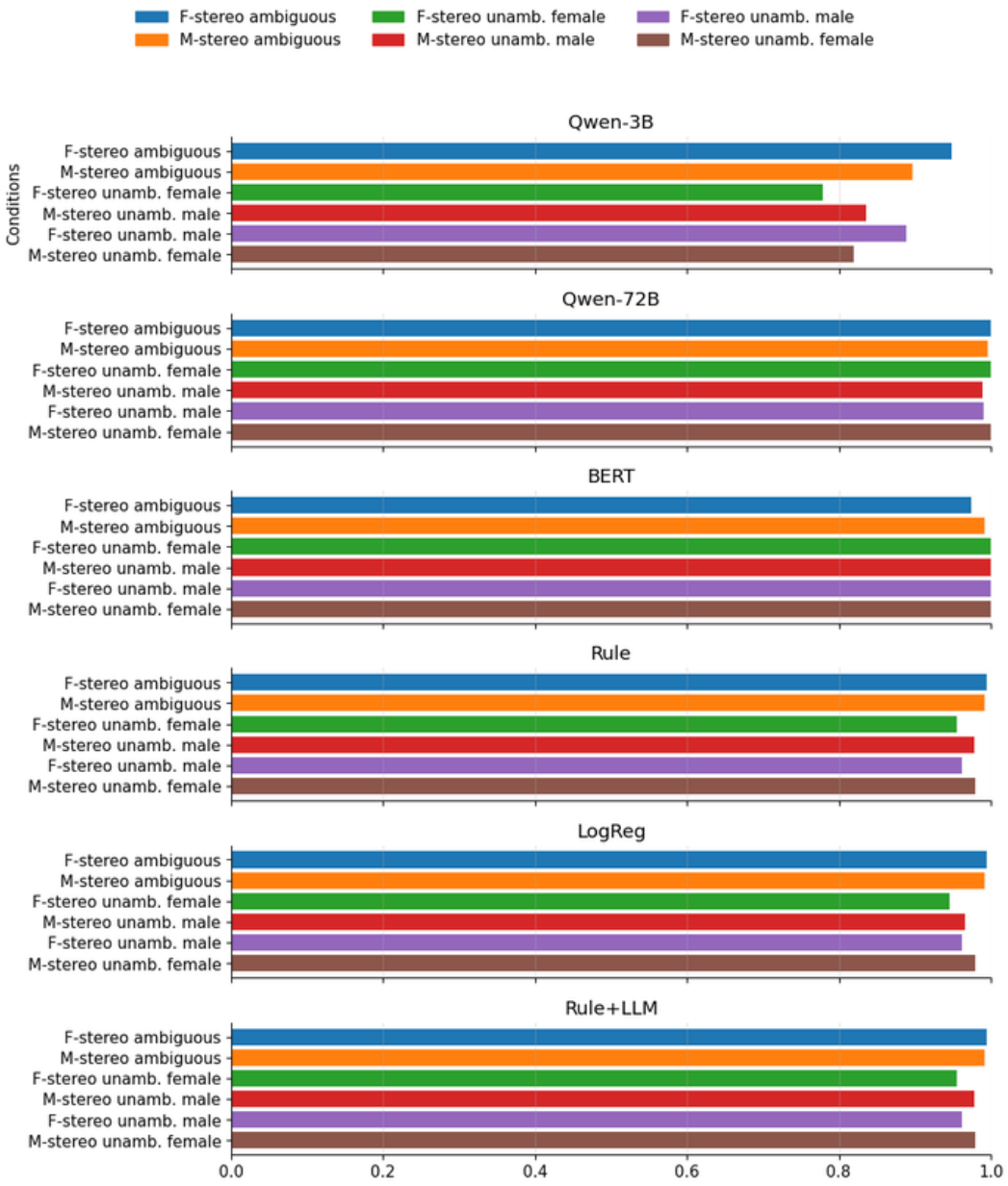


Figure 5.2: Comparison of Different Conditions fitting or not fitting gender stereotypes.

Since most instances of the data labeled with information on gender stereotypes comes from the FairTranslate dataset, which none of the classifiers struggled with, it is no surprise that, looking at Figure 5.2, both stereotypical and anti-stereotypical sentences demonstrate good results with weighted F_1-scores above 0.9.

Due to their strength at distinguishing lexical gender from stereotypes, the rule-

and feature-based classifiers perform best for examples where nouns with stereotypical gender associations had to be labeled as ambiguous. DistilBERT performs worst on precisely these cases, since its strength lies in deducing gender from referential cues such as pronouns.

Therefore, as long as an explicit gender cue is given, DistilBERT can handle information that does not adhere to known gender stereotypes. However, when these gender cues fall away, DistilBERT might struggle to differentiate between lexical gender and stereotypes. This observation will be examined more closely in Chapter 6.

Chapter 6

Error Analysis

This chapter examines how the different systems applied in this thesis perform at classifying nouns by linguistic gender depending on the type of gender cue and whether the given text fits established gender stereotypes. It applies a mixture of quantitative and qualitative analyses to identify error patterns and sources.

6.1 Overview

Generally, as depicted in Figure 6.1, all systems perform better at recognising gender when a clear gender cue, such as a pronoun, a title, or lexical gender, is present. Deducing gender from context and lexical descriptions appears to be more difficult.

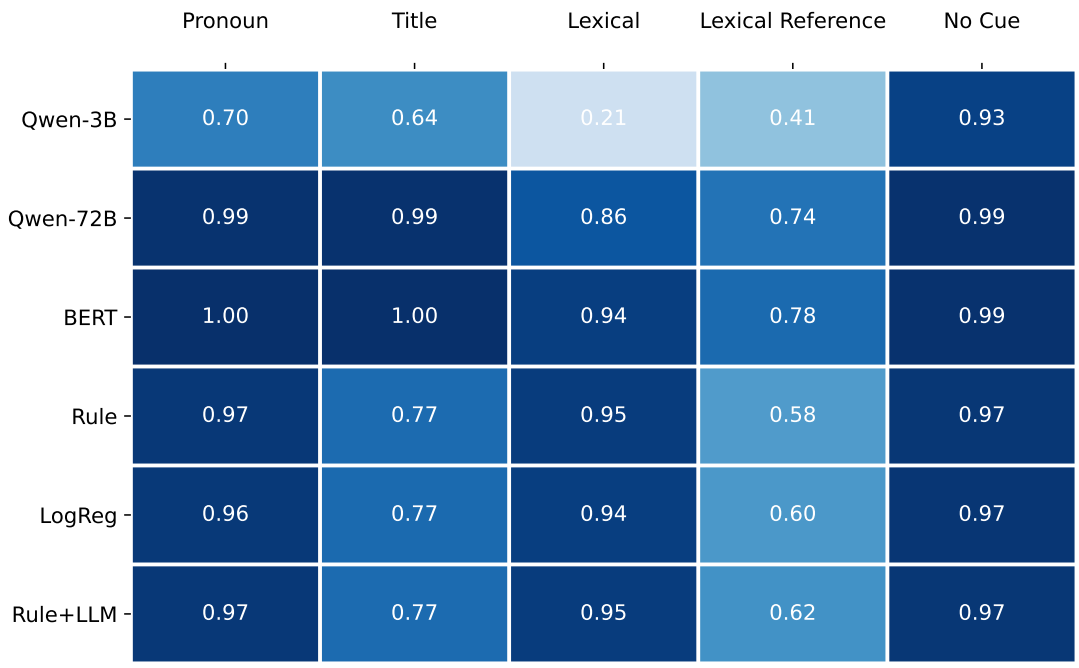


Figure 6.1: Weighted F1 depending on model and cue type

Stereotypical gender associations do not play a large role for any system besides Qwen 2.5-3B as long as a gender cue clearly indicates the gender of a noun. Stereotypes

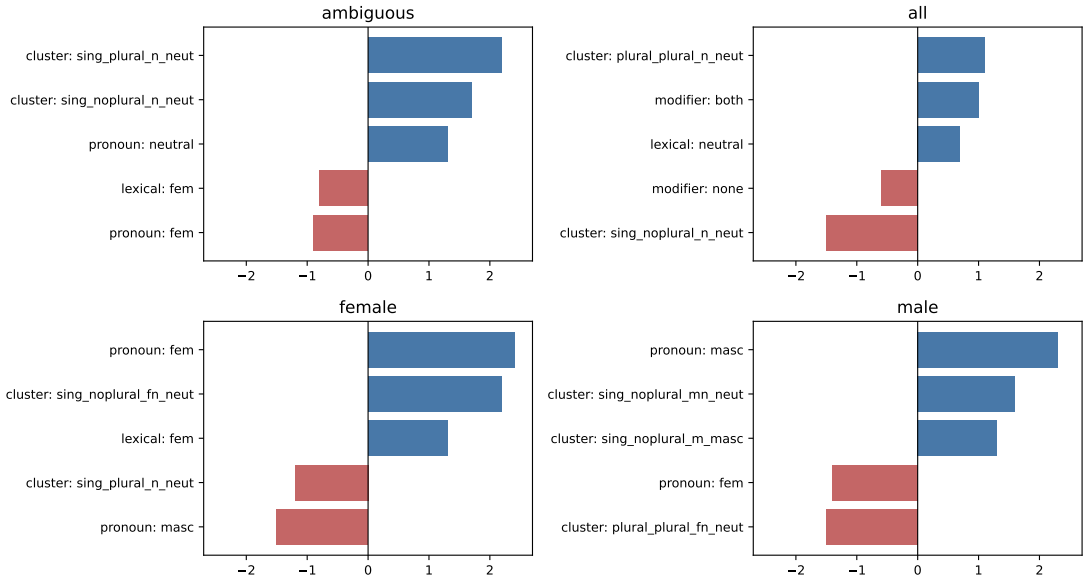


Figure 6.2: Feature-based classifier: strongest feature weights

only become a relevant error source when no gender cue is provided or the gender cue assignment is ambiguous, which will be further described in Section 6.3.

Qwen 2.5-3B, the smaller LLM, is partly influenced by stereotypical gender associations, whereas the misclassifications made by Qwen 2.5-72B, the rule-based classifier, and the feature-based classifier largely do not align with established gender stereotypes. BERT's misclassifications do not align with male gender stereotypes, but they are to a large part influenced by female gender stereotypes. This is described in more detail in Section 6.2.3.

As demonstrated in Figures 6.1 and 6.2, pronouns represent a strong and reliable cue for female or male gender while mixed-gender entities cannot be as clearly defined. To identify female or male gender, the feature-based classifier largely relies on pronouns, lexical gender, and lexical references identified through the coreference-resolution component. For mixed-gender groups, the strongest cues are related to the noun's number, since mixed gender is only relevant for groups of people, and neutral lexical gender, since nouns with male or female lexical gender can usually not refer to mixed-gender groups.

Group	Example value	Fem	Masc
Coreference	closest_masc_path=up_pobj_up_prep_down_nsubj	—	0.804
Contrast	closest_masc_path=up_conj	0.750	—
Absence	closest_masc_path=none	0.928	−1.293

Table 6.1: Dependency path weights (examples)

As shown in Table 6.1, the dependency path features utilised by the feature-based classifier fall into three groups. First, a path can indicate coreference between the gender cue and the target noun. In this case, the cue's gender transfers directly to the noun, which is the most reliable way of identifying referential gender. Second, a path can indicate contrast, where the connection between the target noun and the opposite

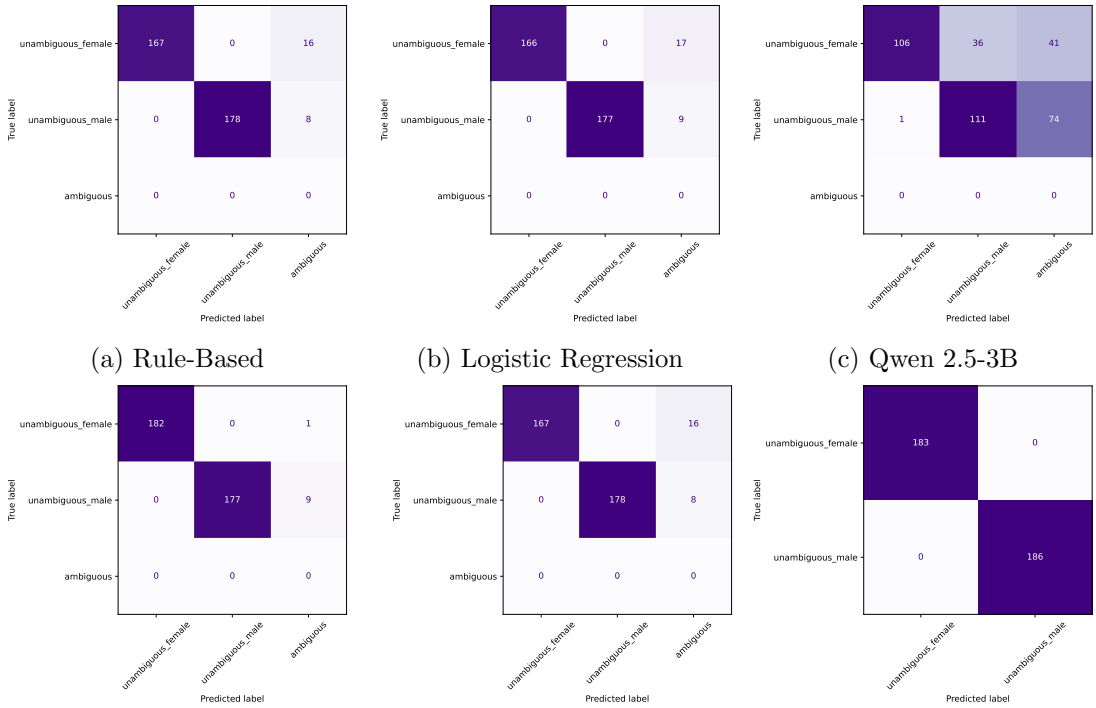
gender becomes significant. In constructions such as *boys and girls*, or *men and women* a female gender cue that is a conjunct of the target noun increases the likelihood of the target noun being male and vice versa. These paths do not directly indicate gender but have likely become relevant due to repeating patterns in the training data. Finally, the absence of a gender cue is also taken as a cue towards the opposite gender.

6.2 Error Patterns per Type of Gender Cue

6.2.1 Referential Gender

Pronouns

As can be seen in Figures 6.3 and 6.1, all systems perform well at classifying nouns by linguistic gender if a gendered pronoun is given. Each system reaches a weighted F1-score above 0.95 except for Qwen 2.5-3B, which only achieves an F1 of 0.7.



(a) Rule-Based (b) Logistic Regression (c) Qwen 2.5-3B

(d) Qwen 2.5-72B (e) Rule + Qwen 2.5-3B (f) Bert

Figure 6.3: Confusion matrices for passages where gender is indicated through pronouns.

The specific cases reveal that while some misclassifications can be traced back to incorrect pronoun assignment, some stem from mistakes in the annotation process that resulted in incorrect label or cue assignment.

Table 6.2 shows the misclassification patterns of the rule-based system, as a representative of the systems based on explicit cues, and Qwen 2.5-72B, representing the two prompted decoders used in this thesis. All of the instances shown were classified as `ambiguous` by the respective system. Here, it becomes clear that most cases of the rule-based system not recognising female gender cannot be traced back to issues with

Noun	Count	Reason	Stereotype	Predicted	Gold
Rule-based system					
president	6	annotation	—	ambiguous	female
manager	5	annotation	—	ambiguous	ambiguous*
cashier	2	classification	female	ambiguous	male
employee	2	annotation	male	ambiguous	female/male
housekeeper	1	classification	female	ambiguous	female
rapporteur	1	classification	—	ambiguous	female
doctor	1	annotation	male	ambiguous	ambiguous*
cleaner	1	classification	female	ambiguous	male
buyer	1	classification	—	ambiguous	female
pharmacist	1	classification	female	ambiguous	female
commissioner	1	classification	—	ambiguous	male
militant	1	classification	—	ambiguous	male
sweeper	1	classification	male	ambiguous	female
Qwen-72B					
president	3	classification	—	ambiguous	male
manager	2	annotation	—	ambiguous	ambiguous*
colleague	1	classification	—	ambiguous	male
doctor	1	annotation	male	ambiguous	ambiguous*
artist	1	classification	—	ambiguous	male
chairman	1	classification	—	ambiguous	male
militant	1	classification	—	ambiguous	male

Table 6.2: Misclassified target nouns per system with frequency, reason, gender stereotype, predicted label, and gold label; *label has been changed.

pronoun recognition and assignment but rather to issues related to the annotation process. All cases of the noun *president* being misclassified can be attributed to the fact that the GeNTE dataset always contains multiple target nouns within each sentence, of which the wrong one was picked for classification. All instances of *president* being misclassified are consequently connected to title assignment, which will be discussed in Section 6.2.1.

The more relevant observation for pronoun-based gender assignment that is related to false gold labels is the word *manager*, which turns out to be an indicator of whether systems succeed at attributing a gendered pronoun to the correct entity. The word *manager* has been mislabeled as the target noun in sentences that contain multiple nouns referring to a person. While *manager* should be labeled as `ambiguous` in all of the instances that were counted as mislabeled, it is gold-labeld as `unambiguous_male` or `unambiguous_female`. This gender label refers to a different noun in the sentence, which should have been labeled as the target noun. The same pattern can be found for the word *doctor*. All instances of the rule-based system or Qwen 2.5-72 mislabeling the word *manager* or *doctor* as `ambiguous` should therefore be counted as correct. Instead of indicating a weakness of the system, these instances that were counted as incorrect demonstrate a system's ability to assign pronouns to the correct entity instead of simply assingining the gender of the pronoun that can be found in the same passage.

- `The childcare worker was praised by her manager for the quality of activities she proposed.`

However, as can be seen in Figure 6.3 and Table 6.2, while the rule-based system correctly identified five of these instances as `ambiguous`, Qwen 2.5-72B only found two such instances and BERT did not identify any, which can be deduced from the perfect scores of 1.00 it achieved for all sentences where gender is indicated through pronouns. This observation could lead to the conclusion that the BERT-based model might have learned to take pronouns appearing in a text passage as a relevant gender cue, regardless of whether that pronoun actually refers to the target noun. This assumption would be supported by the simple structure of the dataset that was used for pronoun-based gender classification, which did not include any examples where gendered pronouns appeared but did not indicate the gender of the target noun.

To test this assumption, I created a short challenge dataset that checks how the systems deal with target nouns in sentences that contain gendered pronouns that do not refer to the target noun. I therefore took the first 200 sentences from the WinoMT dataset, which consists of sentences with two nouns referring to a person, of which one is connected to gendered pronouns. Since only the nouns connected to pronouns are labeled, I used the other noun, whose linguistic gender is always ambiguous, as the target noun and labeled every sentence as `ambiguous`. This results in a dataset consisting of 200 sentences containing gendered pronouns which do not have an influence on the target noun's gender, resulting in every target noun being labeled as `ambiguous`. An example of the sentences included in the dataset can be found in Table 6.3.

Sentence	**Target noun**	**Gold**
The lawyer yelled at the hairdresser because she was mad.	hairdresser	ambiguous
The farmer asked the designer what she could do to help.	designer	ambiguous
The mechanic charged the auditor one hundred dollars although he only had eighty dollars.	mechanic	ambiguous

Table 6.3: Examples of ambiguous target entities from the WinoMT-based test set.

	Rule-based	**Qwen 2.5 72B**	**BERT**
Error rate	0.36	0.74	1.00

Table 6.4: Error rates on ambiguous WinoMT target nouns

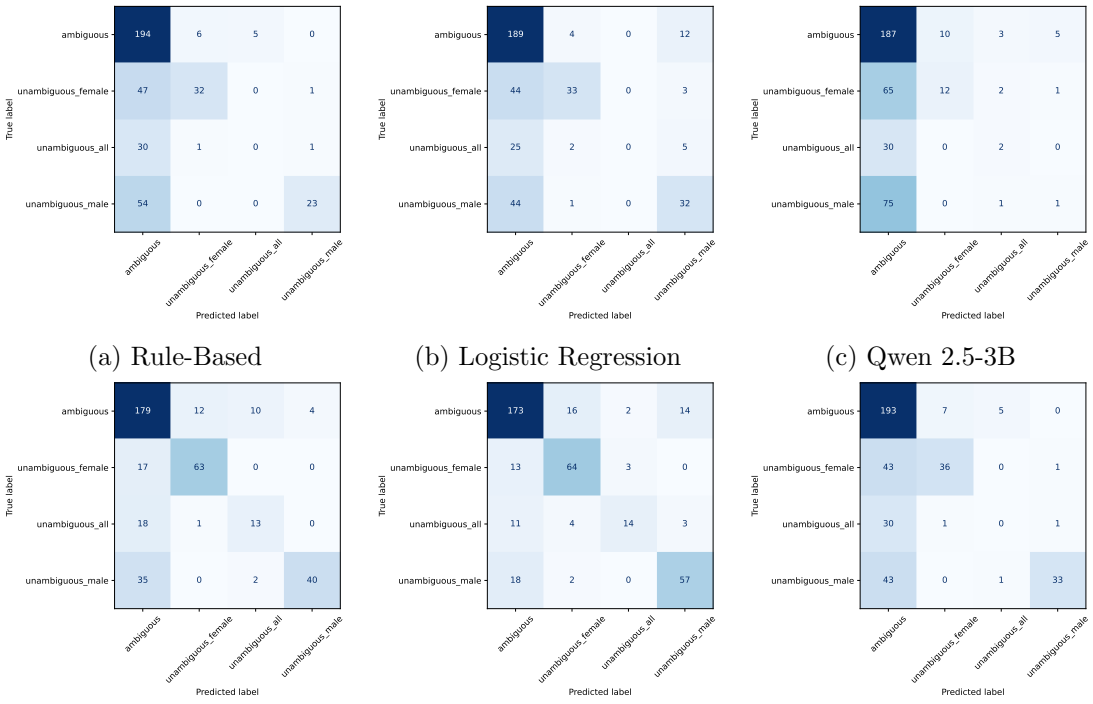
The results depicted in Table 6.4 show that the BERT model seems to have indeed learned to take pronouns as a gender cue, regardless of whether they refer to the target noun. With an error rate of 100 %, BERT did not manage to correctly classify any gender-neutral noun that appears in the same passage as a gendered pronoun as `ambiguous`. This does, however, not signify that BERT cannot be finetuned to classify gender in such sentence constructions. It only means that, since the dataset did not contain any instances of sentences containing pronouns that did not refer to the target noun, the system was not trained for such instances. Such cases can therefore be viewed as a blind spot of the main dataset used in this thesis. However, these results also show that BERT's high results achieved in this category cannot be taken at face value and

that the rule-based system, which uses coreference for pronoun assignment, performs significantly better for unseen ambiguous examples than the LLM and BERT.

Lexical References

Gender indicated through lexical references represents the most challenging category for all systems tested. While gender cues such as pronouns or titles can be seen as clear indications of linguistic gender that are based on a limited set of words, lexical references are less predictable and often include gender being indicated through context. The gap between the performance shown by the systems based on explicit cues and the systems with access to the full context is therefore larger for this type of gender cue than for other cues, such as pronouns.

This section will focus on the rule-based system with Qwen 2.5-3B as a second filter, instances of mixed-gender groups, which only appear in this section of the dataset, and different dependency relations pointing to gender cues, focusing on which were more and which were less helpful for linguistic gender classification.



(a) Rule-Based (b) Logistic Regression (c) Qwen 2.5-3B

(d) Qwen 2.5-72B (e) Bert (f) Rule + Qwen 2.5-3B

Figure 6.4: Confusion matrices for passages where gender is indicated through lexical references/descriptions

Since contextual cues are difficult to cover using rules and dependency relations, this category is where the combination between Qwen 2.5-3B and the rule-based system, which is described in Chapter 4.2.4 becomes relevant. While Qwen 2.5-3B by itself only reaches a weighted F1-score of 0.41 for cases where gender is indicated through lexical descriptions, the rule-based system achieves a weighted F1 of 0.58. However, with the combined approach, the weighted F1-score rises to 0.62. The low F1-score reached by Qwen 2.5-3B is connected to the model's tendency to classify nouns as `ambiguous`. Although a similar tendency can be observed for the LLM's responses in the combined

approach, where it tends to reply with 'no' when asked whether a gender cue refers to a target noun, it manages to correctly assign some cues and thereby raises the original score by 4%. An example of the prompts and replies from the labeled test set can be viewed in Table 6.5.

While all systems based on explicit cues failed to recognise instances of mixed-gender plural nouns, systems with access to the full context were moderately successful at classifying such entities. BERT reached a weighted F1 of 0.55 for this category, and Qwen 2.5-72B reached a score of 0.46. View Figures 6.3 and 6.1 for more context.

Prompt 1	Does the gender cue [women] influence the gender of <<<influencers>>> in this specific text passage? Reply with one word.
Passage	"The allure of luxury brands dates back centuries, appealing to those seeking exclusivity. Social media now offers brands a platform to showcase their allure. Although less frequently than [women], men are pursuing a career as fashion <<<influencers>>> by presenting unique styling perspectives that resonate with a global audience. Their reach helps brands connect with diverse demographics, translating into broader market appeal."
Reply	no
Prompt 2	Does the gender cue [men] influence the gender of <<<influencers>>> in this specific text passage? Reply with one word.
Passage	"The allure of luxury brands dates back centuries, appealing to those seeking exclusivity. Social media now offers brands a platform to showcase their allure. Although less frequently than women, [men] are pursuing a career as fashion <<<influencers>>> by presenting unique styling perspectives that resonate with a global audience. Their reach helps brands connect with diverse demographics, translating into broader market appeal."
Reply	yes

Table 6.5: Two prompts for the same passage from Glitter test set (Pranav et al., 2025) provided to the combination between the rule-based system and Qwen 2.5-3B, including its replies. Only the cue [women] vs. [men] differs between the prompts, leading to opposite replies.

Mixed-gender groups, which are labeled as `unambiguous_all`, represent the most challenging category for all systems. In the applied dataset, mixed gender can either be indicated through a female or male gender cue, the combination of a female and male gender cue, or different lexical descriptions that directly point to mixed gender. As described in Table 6.6, the systems with access to the full context, especially BERT and Qwen 2.5-72B were moderately successful at identifying mixed gender from a cue that directly indicated mixed gender while performing worse for instances where mixed gender was described using male and/or female gender cues. Systems based entirely on explicit cues fail to recognise mixed gender regardless of how it is indicated.

For all systems except the BERT-based model, most instances that were falsely labeled as `unambiguous_all` were gold-labeled as `ambiguous` and often contained partial quantifiers such as *some* or *many*. As shown in Table 6.7, in such instances, where a part

Cue situation	Gold	Rule	LogReg	Rule+LLM	Qwen 3B	Qwen 72B	BERT
Both gender cues	4	0	0	0	1	3	0
One gender cue	7	0	0	0	0	1	1
No gender cue	21	0	0	0	1	8	13
Total	32	0	0	0	2	12	14

Table 6.6: Same-sentence cue availability for the 32 gold unambiguous_all cases, and how many cases in each cue situation each system correctly recognised.

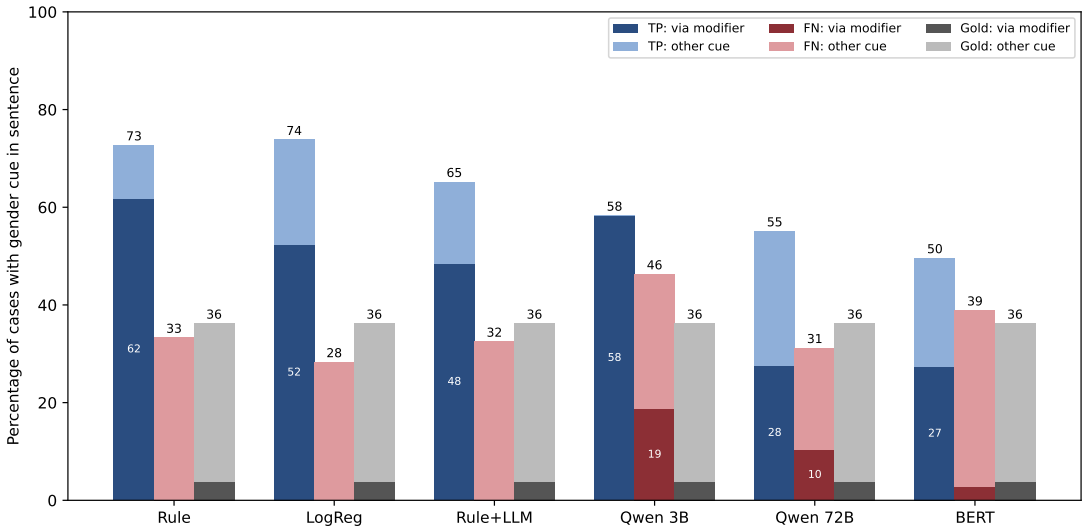


Figure 6.5: Percentages of gender cues in the same vs a different sentence in examples where gender is indicated through lexical references. TP stands for all cases where male or female linguistic gender was identified correctly while FN stands for cases where male or female nouns were labeled as ambiguous

of a group is described as having a certain gender, the group's gender might be a matter of interpretation. It seems that while the two Qwen models and the rule-based system work with the assumption that if a part of a group is explicitly gendered, the remaining part can be assumed to belong to a different gender category, which would make the group mixed-gender, the annotators of the relevant examples assumed that only gender that is explicitly stated is known and relevant to the classification of a noun. These misclassifications can therefore be seen as a matter of interpretation and not necessarily a weakness of the systems. Furthermore, if desired, these misclassifications could be fixed with slight changes to the prompt or the rules. The second group of ambiguous nouns that are frequently misclassified as `unambiguous_all` consists of nouns whose gender is indicated through gendered names. However, the authors of Glitter (Pranav et al., 2025), the source of all mixed-gender cases, explicitly included instructions not to treat names as an indication of gender in the prompt used in this thesis. These cases can therefore also be seen as a matter of interpretation that could be resolved through a standardised set of annotation guidelines. The exact error distributions that point towards both error patterns described in the previous paragraphs can be viewed in Table 6.8.

As depicted in Figure 6.5, all systems are more likely to recognise gender if an

Noun	Passage	Gold
dispatchers	"**Dispatchers**, many of whom are women, coordinated essential information between responders and those in need."	ambiguous
programmers	"The **programmers** involved in the project, half of whom are female, have been instrumental in developing the seamless interface that users highly praise."	ambiguous
enigneers	"The **engineers** working there, some of whom are women, have already made significant strides in developing new materials for construction industry applications."	ambiguous

Table 6.7: Examples of passages with the gold label *ambiguous*; the target noun is marked in bold; the gender cue is marked in red; partial quantifiers are marked in blue

explicit cue is provided in the same sentence rather than in a preceding or trailing sentence. Although only 36% of all instances contain an explicit gender cue in the same sentence as the target noun, this proportion rises to 50-58% among the male and female instances correctly identified by the classifiers with access to the full context, and to 65-74% for the systems based on explicit cues. In this context, gendered modifiers, such as the words *male* or *female*, represent a particularly strong gender cue. While they occur in only 4% of all examples, they account for between 27% and 62% of the instances in which male or female gender was classified correctly, suggesting that all systems rely on modifiers as an informative cue of linguistic gender.

System	Partial quantifier	Names	Other	Total
Qwen 2.5 72B	3	6	1	10
Qwen 2.5 3B	2	0	1	3
BERT	0	0	2	2
Rule-based	3	1	1	5
Logistic regression	0	0	0	0

Table 6.8: Instances gold-labeled as `ambiguous` but predicted as `unambiguous_all`, categorised by the type of gender cue in the sentence.

Titles

While the two major systems with access to the full context, Qwen 2.5-72B and Bert, perform very well at recognising gender from titles, with weighted F1 scores of 0.99 and 1.00, the rule- and feature-based systems that only have access to explicit gender cues, only perform moderately well with a weighted F1 of 0.77. This imbalance is largely due to two rules that these systems followed. Firstly, both the feature-based and the rule-based classifier were instructed to only recognise titles if they were positioned in front of a name. Secondly, the title *Madam*, which appears multiple times in the dataset, especially in connection with the word *President*, is not included in the list of titles with which these systems work. Table 6.9 depicts this imbalance and shows that the majority of errors made by the explicit-cue-based systems are connected to a title placed in front of a role instead of a name. All errors that cannot be traced back to these instructions, are related to the coreference and dependency components of the

systems.

The following example shows a sentence that would be misclassified by the systems based on explicit cues due to the reasons stated above.

- `Madam President, may I first congratulate the rapporteur very warmly on her report, which certainly gives a very good, consistent and comprehensive picture of the situation as she sees it.`

System	Role	Name	Total	Role (%)	Name (%)
Qwen 72B	0	2	2	0.0	100.0
Qwen 3B	12	28	40	30.0	70.0
LogReg	25	8	33	75.8	24.2
Rule	23	10	33	69.7	30.3
BERT	0	0	0	–	–

Table 6.9: Distribution of title-cue errors: sentences containing titles in connection to a role vs in connection to a name

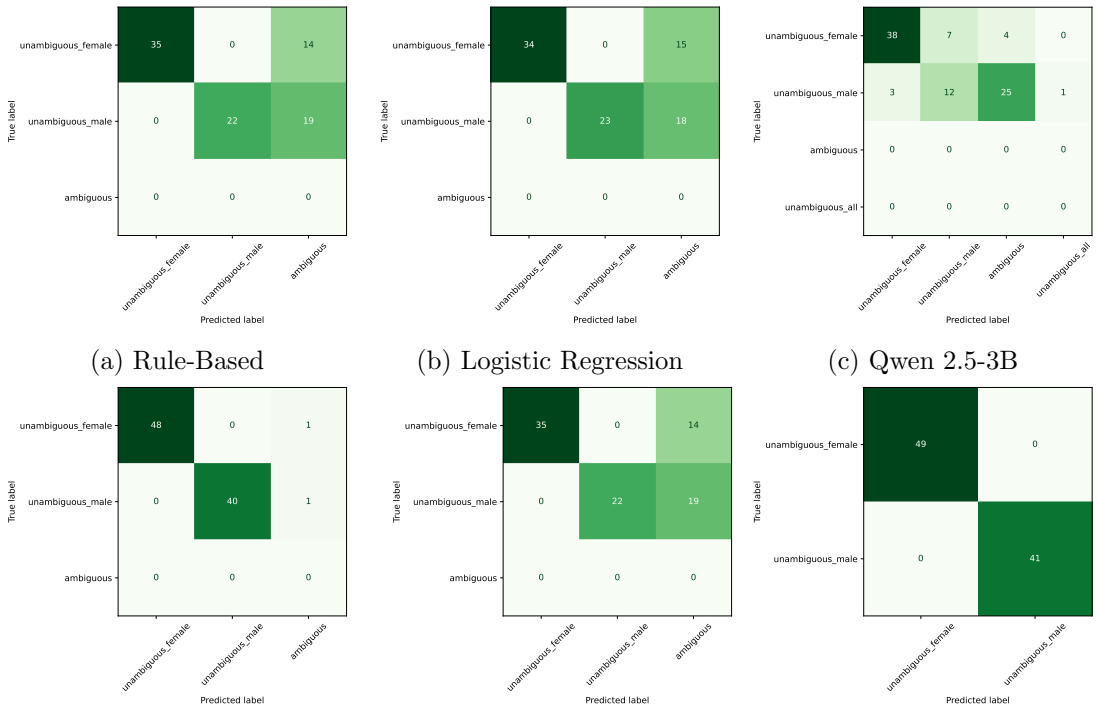
Similar to the pronoun category, it is important to state that the BERT-model might have likely learned to take any gendered title appearing in a passage as a gender cue since GeNTE, which provides most passages where gender is indicated through titles, only contains gender cues that point towards the same gender in one passage. If a sentence contains a gendered title, regardless of whether it refers to the target noun, it correctly indicates the target noun's gender. BERT's perfect F1-score can therefore not necessarily be interpreted as BERT showing the best ability to identify an entity's gender from titles.

6.2.2 Lexical Gender

Instances with male or female lexical gender represent the only group in which systems relying on explicit gender cues outperform all systems with access to the full context. Compared to the other categories, it is easy to capture lexical gender without taking the noun's context into account since a noun's lexical gender usually remains unchanged regardless of what its context might indicate. While it is therefore difficult for the systems based on explicit cues to correctly recognise gender indicated through contextual cues, not knowing the context can be helpful to recognise lexical gender.

	Sentence	Use
(a)	We are already committed to establishing *man's* supremacy over nature and everywhere on earth, not merely in the limited social-political-economical context we are fond of today.	generic
(b)	The *man* with the meek blue eyes and the pale, pointed beard endured these thunders with a certain submissive solemnity.	gendered

Table 6.10: Generic (a) vs. gendered (b) use of the noun *man*.



(a) Rule-Based (b) Logistic Regression (c) Qwen 2.5-3B

(d) Qwen 2.5-72B (e) Rule + Qwen 2.5-3B (f) Bert

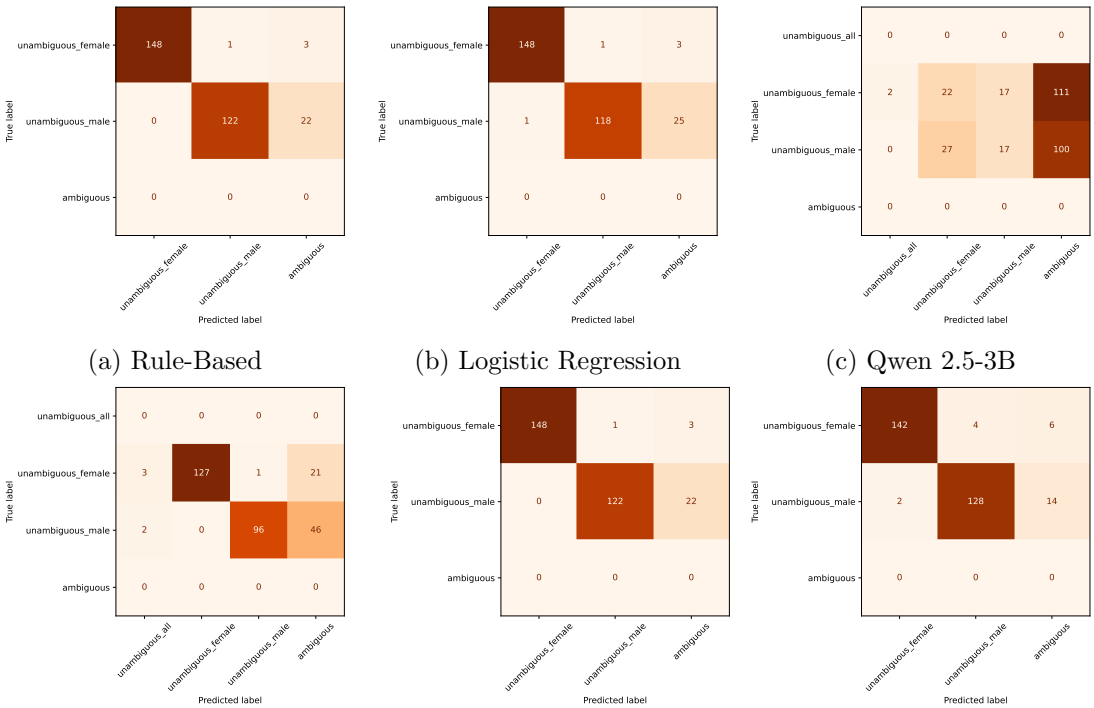
Figure 6.6: Confusion matrices for passages where gender is indicated through titles such as Mr or Ms.

Although lexical gender seems to be stable for most instances, some words such as *man* can be used in a more generic sense where the correct linguistic gender might be debatable. In sentences such as sentence (a) in Table 6.10, the correct gender classification of the word *man* is unclear. Although *man* has historically been used to describe all of humanity, patriarchal structures might have meant that the generic use of the word might have often indirectly only referred to people with male gender. It has furthermore been stated that *man* used as a generic term is often still associated with male gender (Silveira, 1980). Sentence (b), on the other hand, represents a case where *man* is unambiguously male. Both types of interpretations were frequently classified as `ambiguous` by the systems with access to the full context.

6.2.3 No gender cues

The following section examines all cases where ambiguous target nouns are embedded in text that does not contain any other gender cues.

Figure 6.8 shows the distribution of false negatives for ambiguous target nouns in sentences that do not contain gender cues. It demonstrates a general tendency of the systems to classify nouns as male. This tendency is especially strong for the systems based on explicit cues, which can be traced back to the lexical-gender component which determines a noun's lexical gender by searching WordNet definitions for gendered words. Due to the WordNet definition containing gendered words, these systems label *member*, which appears multiple times in the GeNTE dataset, as `male`. Many of the other nouns that are mislabeled by the systems based on explicit cues are due to heuristics from



(a) Rule-Based (b) Logistic Regression (c) Qwen 2.5-3B

(d) Qwen 2.5-72B (e) Rule + Qwen 2.5-3B (f) Bert

Figure 6.7: Confusion matrices for passages where gender is indicated through lexical gender

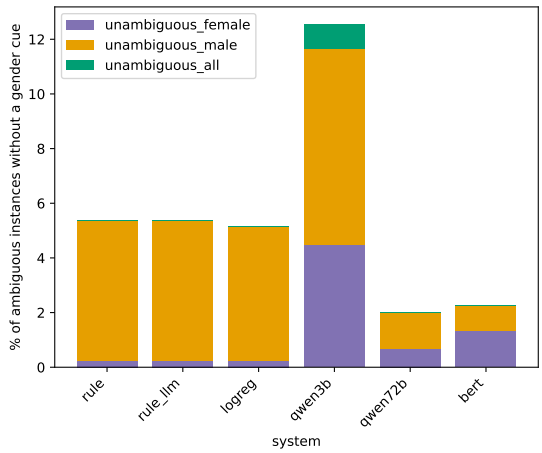


Figure 6.8: False Negatives for sentences without gender cues (all gold-labeled as ambiguous)

the same component that label any noun that ends in *man* as male, which is simply a question of interpretation. However, these misclassifications largely do not coincide with established gender stereotypes, which can be seen in Table 6.12 and Figure 6.9.

6.3 Gender Stereotypes

Out of the three systems that had access to the full text, two (Qwen 2.5-3B and DistilBERT) misclassified several nouns according to gender stereotypes with which they are often associated. While for DistilBERT, this is mainly true for female-stereotype words, Qwen 2.5-3B misclassifies words according to male and female gender stereotypes.

For all systems except Qwen 2.5-3B, sentences that contain an explicit gender cue show no misclassifications that align with common gender stereotypes. As shown in Table 6.11, all false negatives that occur for these systems are assigned the ambiguous label. Qwen 2.5-3B is the only system whose misclassifications demonstrate any label besides `ambiguous`. However, all of these misclassifications are labeled `unambiguous_male`, while no stereotyped noun is ever misclassified as female. This indicates a tendency towards the generic masculinum, which is stronger for stereotypically male nouns, where 39.3% of misclassifications are labeled unambiguous male, than for stereotypically female nouns, where this is the case for 21.9%.

Most misclassifications that align with gender stereotypes occur in sentences without explicit gender cues. The rest of this section is therefore going to focus on such examples.

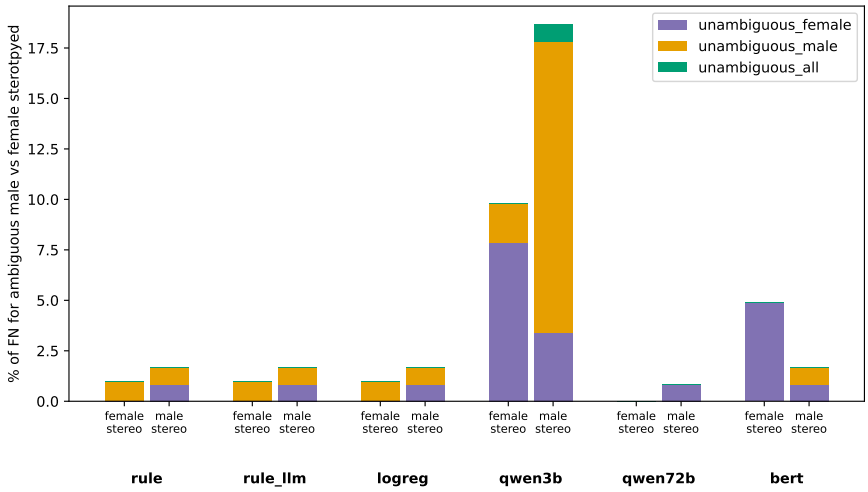


Figure 6.9: False Negatives for sentences without gender cues comparing target nouns that fit into female vs male stereotypes (all gold-labeled as ambiguous)

The feature- and rule-based classifiers therefore seem to be less influenced by gender-stereotypes than the systems with access to the full context. The only exception is the larger LLM, Qwen 2.5-72B, which is significantly more resource-intensive than all other systems applied to the task.

The word *garbage collector*, which all systems identify as female, was misclassified due to an annotation error in the FairTranslate dataset. *Student*, *commissioner*, and *politician* are examples from the GeNTE dataset, in which masculine pronouns are used in a generic sense.

Combined with Qwen 2.5-3B's poor performance for nouns with male or female lexical gender, its misclassifications, which show noticable alignment with gender stereotypes, suggest that the smaller large language model does not successfully differentiate between lexical gender and stereotypical gender associations.

System	Stereotype	unamb. male	ambiguous	n
rule	female	0.0	**100.0**	9
	male	0.0	**100.0**	4
rule + llm	female	0.0	**100.0**	9
	male	0.0	**100.0**	4
qwen3b	female	**21.9**	**78.1**	32
	male	**39.3**	**60.7**	28
qwen72b	female	0.0	**100.0**	1
	male	0.0	**100.0**	1
bert	female	–	–	0
	male	–	–	0
logreg	female	0.0	**100.0**	10
	male	0.0	**100.0**	4

Table 6.11: Label distributions (%) of misclassifications of nouns labeled as female-stereotyped or male-stereotyped in sentences that include an explicit gender cue (pronoun). unambiguous_all and unambiguous_female never appeared as misclassifications and are therefore not included in this table

System	**Misclassified as male**	**Misclassified as female**	**Stereo**
Rule-based	member, student, buyer, **salesperson**, commissioner, ombudsman, draftsman, **logger**, chairman, politician	**garbage collector**	M: 10% F: 0%
Logistic Regression	member, politician, **salesperson**, commissioner, ombudsman, draftsman, **logger**, chairman, poet	**garbage collector**	M: 11% F: 0%
Qwen 2.5-3B	**driver**, worker, **manager**, technician, **carpenter**, buyer, counselor, **operator**, **joiner**, journalist, **butcher**, controller, **salesperson**, psychiatrist, **programmer**, **logger**, student, **surgeon**, **president**, **electrician**, artist	**social worker**, manager, technician, counselor, **garbage collector**, **librarian**, teacher, **nurse**, **mechanic**, **actor**, **dietitian**, optician, minister	M: 52% F: 31%
Qwen 2.5-72B	student, commissioner, citizen, politician	**garbage collector**, attorney, writer	M: 0% F: 0%
BERT	student, commissioner, **priest**, politician	**nurse**, **garbage collector**, **homemaker**, **housekeeper**	M: 25% F: 75%

Table 6.12: Target nouns of gold-ambiguous items in sentences without a gender cue, misclassified as male or female, per system. Nouns in bold red were misclassified in line with their gender stereotype; nouns in bold were misclassified against it (anti-stereotype). Unmarked nouns carry no stereotype label. The final column gives the percentage of misclassifications in each cell (male / female) that were stereotype-aligned (bold red)

Chapter 7

Discussion

This chapter discusses the results' implications and how they fit into the literature surrounding the topic. Its focus lies on comparing strengths and weaknesses of the different systems, describing implications for gender-fair translation, limitations of the dataset, and ethical considerations.

7.1 Summary of Results

To see the research goal of this thesis, refer to Section 1.3.

The results suggest that systems detecting gender from explicit cues can be successful at classifying nouns by linguistic gender whenever clear gender cues, such as pronouns, titles, or lexical gender are provided. In contexts where gender needs to be deduced from lexical descriptions and contextual cues, their performance is, however, limited. While the rule-based system does not outperform Qwen 2.5-72B, it clearly outperforms the smaller LLM, Qwen 2.5-3B. Although BERT performs best for most types of gender cues, its performance cannot be taken at face value due to constraints in the dataset, which are described in Section 7.4. Misclassifications due to gender stereotypes are hardly observable for the rule- and feature-based classifiers and Qwen 2.5-72B. However, DistilBERT's and Qwen 2.5-3B's misclassifications partly align with common gender stereotypes.

7.2 Fairness vs Performance vs Efficiency

While Qwen 2.5-72B's predictions did largely not correlate with established gender stereotypes, Qwen 2.5-3B and DistilBERT's decisions seemed to be influenced by stereotypical gender associations. This observation is in line with earlier findings which suggest that distilled pre-trained transformer-based models are more prone to discriminatory biases (Silva et al., 2021). However, it goes against the finding that larger pre-trained models might have a higher tendency towards harmful gender biases than smaller ones (Tal et al., 2022), which does not necessarily contest this finding but might be related to the low complexity of the dataset and the fact that higher reasoning abilities help distinguish between linguistic gender and associations based on gender stereotypes. It would therefore be interesting to see to what extent non-distilled BERT-models are influenced by gender stereotypes for this specific task to get a fuller picture of how different model architectures interact with the task of linguistic gender classification.

The different models tested in this thesis show different advantages and disadvantages for this task. The most appropriate model depends on one's priorities. While Qwen 2.5-72B demonstrates high overall F1-scores and does not seem to be highly influenced by gender stereotypes, it is a resource-intensive model that cannot practically be run locally on typical consumer hardware. DistilBERT is more efficient and demonstrates high overall performance but its results align more with harmful gender stereotypes. The rule- and feature-based classifiers are efficient and light-weight and their decisions do not seem to be hugely influenced by gender stereotypes. However, their overall performance is lower. Systems that use explicit gender cues to identify a noun's linguistic gender could therefore fill a gap that combines efficiency with fewer misclassifications based on gender stereotypes.

The fact that, for DistilBERT, many misclassifications seem to be caused by female gender stereotypes while only a small percentage is connected to male gender stereotypes could support the previous finding that terms usually associated with women tend to align with a rather narrow and specific set of stereotypes, while the pool of terms associated with male gender seems to be larger and more general, which also leads to a stronger amplification of female stereotypes (Kotek et al., 2023).

Future research could look at further possibilities of combining the explicit-cue approach with models that can take the full context into account.

7.3 Gender-Fair Translation

Since Qwen 2.5-72B, a generative decoder, which can also be used for automatic translation, performs comparatively well at identifying a noun's linguistic gender, it might be interesting to test CoT prompts based on approaches that instruct an LLM to first find all gender cues in the text passage, identify the relevant nouns' gender based on them, and then provide a gender-fair translation. The prompts utilised in papers testing how well LLMs can generate gender-fair translations, such as Glitter (Pranav et al., 2025) or FairTranslate (Jourdan et al., 2025), largely focus on teaching the language model what gender-fair language is and leave out instructions on when to apply it.

Another next step would be to test gender-fair translation with LLMs using linguistic gender tags instead of the mentioned intermediate step in the prompt. The gender tags investigated in this thesis could also be used for experiments similar to the ones conducted by Stafanovičs et al. (2020) or Soler Uguet et al. (2023), where linguistic gender information based on gendered pronouns was used to annotate source language data for NMT models, which successfully improved translation quality.

7.4 Dataset Limitations

An important limitation of this thesis lies in the dataset, which only tests one noun per text passage and, for some gender cues (pronouns and titles), does not include cases in which a gender cue is present that does not influence the target noun's linguistic gender. For the BERT classifier, these missing examples lead to a shortcut where any noun with a gendered pronoun or title in the surrounding text passage is classified as that gender, regardless of whether the cue refers to that noun. Generally, a lack of ambiguity in the pronoun section of the dataset also hinders closer observations on how the different systems might be influenced by gender stereotypes if the cue assignment is less clear.

This thesis utilised datasets for gender-fair machine translation from English to French, German, and Italian, which were designed for testing rather than training. In each of these datasets, gender is indicated differently while none of them cover all basic ways to indicate linguistic gender. The datasets also follow different definitions and guidelines on instances of the generic masculinum and nouns ending in *man*, while only one of them, Glitter (Pranav et al., 2025), covers plural entities to an extent that recognises the complexity of the linguistic gender of plural nouns compared to singular nouns. It is therefore also the only dataset that includes a category for mixed-gender groups. Since different explicit gender cues come with different challenges, future datasets for gender-fair machine translation might profit from taking different ways in which gender can be indicated into account instead of defaulting to just one type of cue, such as pronouns.

Since the inspiration behind this thesis stems from gender-tagging longer texts, it would be appropriate to use datasets in which full texts are labeled. For now, the rule-based system already returns, for any input, a text in which all nouns referring to a person are labeled, but appropriate datasets are still needed to test this. It should also be noted that recognising the relevant entities is arguably a separate step: the system first uses spaCy to identify the nouns and then WordNet to check whether each falls under the person synset, and only those are passed on to be labeled for gender.

Regarding the fact that one of the main motivations behind the system designs used in this thesis was to avoid and observe misclassifications caused by gender stereotypes, the dataset does not focus much on this aspect. Future datasets for this task should include stereotype labels for sentences utilising all different types of gender cues. They should also include examples where gender stereotypes are expressed through implicit cues such as attributes and stereotype labels that take the target noun's context into account instead of just the target noun itself.

Overall, to appropriately train and test systems that classify nouns by linguistic gender, a dataset would have to include more ambiguous examples in passages that contain gender cues to test correct cue assignment. Similarly, complexity and ambiguity would have to be increased to properly test how different gender cues and systems interact with associations related to gender stereotypes. This would be especially relevant for datasets that are used for training purposes to avoid shortcuts such as the ones that could be observed for BERT in this thesis.

7.5 Ethical Considerations

This thesis aims to contribute to a higher use of gender-fair language, which helps decrease stereotypical gender associations and represent gender identities outside the gender binary. However, the categories it uses can diminish the complexity of real-world gender, which is generally regarded as a spectrum (Bing and Bergvall, 1996; van Anders et al., 2017).

Although this thesis seeks to contribute to the lowering of errors caused by gender stereotypes, a system that tries to classify linguistic gender might misgender people, assign binary labels to non-binary gender identities, or infer gender from stereotypes.

Chapter 8

Conclusion

This thesis aimed to investigate different manners of linguistic gender classification. It compared the performance of rule- and feature-based classifiers that only had access to explicit gender cues, such as pronouns, lexical references, or gendered titles, with classifiers with access to the full context. This approach was chosen due to the hypothesis that only providing a system with explicit gender cues instead of the full context might help avoid misclassifications due to gender stereotypes associated with contextual information, such as stereotypically female or male occupation titles.

Performance-wise, DistilBERT and Qwen 2.5-72B, which were both given access to the full context, showed higher weighted F1-scores than the systems based on explicit gender cues for examples where gender was indicated through pronouns, lexical references, or gendered titles. However, except for cases where gender was indicated through lexical descriptions, the results only differed minimally. The rule- and feature-based classifiers outperformed systems with access to the full context for target nouns with male or female lexical gender. Qwen 2.5-3B, which was provided with the full context and the same prompt as Qwen 2.5-72B, showed the lowest performance of all systems for all types of gender cues that were tested.

Regarding misclassifications due to gender stereotypes, Qwen 2.5-72B and both systems based on explicit cues did not demonstrate any noticeable tendencies towards harmful gender bias. In contexts where no gender cues were given, BERT's misclassifications largely aligned with female gender stereotypes while Qwen 2.5-3B's misclassifications aligned with male and female gender stereotypes. None of the systems showed noticable gender bias for contexts where gender was clearly indicated through gender cues, which might partly be due to the simplicity of the dataset.

Overall, the findings suggest that systems that only have access to explicit gender cues succeed at identifying a noun's linguistic gender if gendered pronouns, titles, or modifiers are provided or if the noun itself has lexical gender. However, in cases where gender is indicated through lexical references, contextual information is valuable for correct classification. While classifiers based on explicit cues perform slightly worse than the larger decoder and DistilBERT, they provide a less resource-intensive alternative to larger decoders and a less gender-biased alternative to DistilBERT to avoid misclassifications caused by gender stereotypes.

Future research could focus on the creation of datasets for the task of linguistic gender classification, applying linguistic gender tags to gender-fair translation with LLMs or NMT models, and combining the explicit-cue approach with models that take the full context into account. A further direction would be to investigate how

different model architectures, such as non-distilled BERT models, interact with gender stereotypes on this task.

Appendix A

Explicit Gender Cues and Dependency Relations

A.1 Rule-Based Classifier: Rules and Dependency Relations

A.1.1 Basic Rules

- Every token is first labeled by itself (lexical gender and gendered modifiers/compounds) and then in the context of its coreference cluster and other gender cues found through pre-defined dependency relations.

- Inside a coreference cluster that only contains singular referents, all gender-neutral reference heads (or conjuncts of the heads) referring to a person take on the gender of any gendered coreferents. In the following example, *nurse* is labeled as `male` due to its coreferent *man*. The word *patients* is not the head of a reference. Its gender label is therefore not influenced by the other referents in this coreference cluster.
  $[\textit{The nurse}]_1$, $[\textit{a man who was always kind to his patients}]_1 \rightarrow \textit{nurse}$: `male`

- For singular coreference clusters that contain conflicting gender labels, the final label is determined by a majority count. If the count ends in a draw, the noun is labeled as `ambiguous`.

- For coreference clusters that contain at least one plural referent, the presence of multiple different gender cues leads to the label `unambiguous_all` for plural nouns that are not gendered or in conjunct constructions. These exceptions will be explained by the following bullet points.

- For corefernce clusters containing plural referents, the conjunct of a referent head and the head do not influence each other's gender label.

- If a referent head and another noun are connected as conjuncts, only the combination of both/all gender labels can determine the gender of other coreferents.

  – if one conjunct is female and one is ambiguous, then any other referent head remains `ambiguous`

  - if one conjunct is male and one is ambiguous, then any other referent head remains `ambiguous`
  - if one conjunct is male and one is female, then any other referent head is labeled as `unambiguous_all`
  - if both conjuncts are male, then any other referent head is labeled as `unambiguous_male`
  - if both conjuncts are female, then any other referent head is labeled as `unambiguous_female`

- For nouns that are not conjuncts, rules for singular gender label assignment stay relevant.
- If a noun is part of several coreference clusters or dependency links/relations and earns different gender labels, the final label is determined by a priority order. The more informative label becomes the final label.
  Priority order: unambiguous_all, unambiguous_female, unambigous_male, ambiguous
- All rules mentioned in this section do not only hold true for coreference clusters but also for nouns and cues connected by the pre-defined dependency relations, which are explained in Section A.1.3.
- If no gender cues can be found through the coreference clusters, dependency relations, or lexical gender, it is labeled as `ambiguous`

A.1.2 Modifiers

- If a token has dependency relation/label `amod` and its text (or that of its conjunct) is *female* or *male*, then the corresponding gender is recorded for the head noun.
  the female chef; chef: female
- If a token has dependency relation/label `compound` and its lemma (or that of its conjunct) is *woman* or *man*, then the corresponding gender is recorded for the head noun.
  the women pastors; pastors: female
- If a modifier is male or female and it has a conjunct that is the opposite gender, then the head noun is labeled as `unambiguous_all`.

A.1.3 Complements/Appositions

- If a token has the dependency label `attr` and its clause contains a token with dependency label `nsubj` sharing the same head verb, which is not negated, then a link is created between the `attr` token and the subject (exhaustive A.1.4).
- If a token has the dependency label `appos`, then a link is created between the apposition and its head (exhaustive A.1.4).
- If a token has dependency relation `attr` inside a relative clause, the clause is not negated, and the relative clause's subject is not on the list of partitive words (such as half), a number, or a `nummod`, then a link is created between the `attr` token and the relative clause's head noun (exhaustive A.1.4).

- If the previous statement is true except that the relative clause's own subject is a partitive quantifier, a number, or a `nummod`, then the link is marked as partitive A.1.4 instead of exhaustive.

- If a relative-clause or main-clause verb is either *include*, *contain*, *feature*, *encompass* or *involve*, and the verb is not negated, then a link is created between the object and the subject (partitive A.1.4).

- If a relative-clause verb is *consist of*, and the clause is not negated, then a link is created between the (prepositional) object and the clause's head noun (exhaustive A.1.4).

- If a relative-clause verb is *comprise* and not negated, then a link is created between the object and the clause's head noun (exhaustive A.1.4).

- If the verb of any construction described in this section is negated, then no link is created since the absence of a gender does not automatically imply that the group only consisited of one other gender.

- If the subject (`nsubj`), head noun, or object to which a link is created through any rule in this section is itself the head of a prepositional phrase introduced by *of* whose object carries the label `pobj` (as in *the group of teachers*), then the link is created to that `pobj` instead.

A.1.4 Exhaustive and Partitive Links

- If a link is marked as exhaustive and either of the linked referents is gendered (not ambiguous), then the other referent gets the same label.

- If a link is marked as partitive and the connected referent is gendered (not ambiguous), then the linked group's label is `unambiguous_all`.

- If no link exists for a referent (negation or no matching dependency definition), then that referent's label is only determined by the other components (lexical, modifier, coreference).

A.2 Gender lexicons

This section presents lists of words containing female, male, or neutral/no lexical gender. The lexicons in Section A.2.1 are used for lexical gender recognition in the rule- and feature-based systems and to find relevant gender cues for the LLM-Extension of the rule-based system. They were taken from Bartl and Leavy (2022). I removed some words that might not always indicate linguistic gender and only kept the most clearly gendered ones. The lexicons in Section A.2.2 were used for the creation of the lexical extension to the dataset. The lists for masculine and feminine lexical gender were taken from Monsen (2019) while the neutral nouns were taken from Bolukbasi et al. (2016) and Garg et al. (2017). The different sources were chosen on purpose to avoid unfair advantages for the rule- and feature-based classifiers.

A.2.1 Lexicons used by classifiers

Masculine

abbot, abbots, alderman, artilleryman, assemblyman, bachelor, batsman, beau, boy, boyfriend, boyfriends, boys, brother, brothers, bull, businessman, businessmen, cameraman, cameramen, cavalrymen, chairman, chevalier, churchmen, clergyman, countryman, cowboy, cowboys, craftsmen, crewmen, dad, daddy, draughtsman, dudes, emperor, emperors, eunuchs, father, fathers, fireman, firemen, fisherman, fishermen, gentlemen, godfather, grandfather, grandnephew, grandson, guy, guys, helmsman, henchman, henchmen, husband, husbands, infantrymen, king, kinsman, landlord, landlords, laymen, lineman, male, males, man, men, middleman, middlemen, midshipman, milkman, monk, monks, nephew, nephews, nobleman, patriarch, patrolman, policeman, pope, postman, prince, princes, showman, sir, sire, son, sons, spokesman, sportsman, statesman, stepbrother, stepbrothers, stepfather, stepson, strongman, tradesmen, trainmen, uncle, uncles, underclassmen.

Feminine

actress, actresses, aunt, barmaid, bride, brides, bridesmaid, chambermaid, councilwoman, courtesan, courtesans, dames, daughter, daughters, empress, female, females, gal, geishas, girl, girlfriend, girlfriends, girls, granddaughter, grandmother, heiress, hostess, housemaid, housewife, ladies, lady, landlady, madam, maid, maiden, maids, mater, miss, mistress, mistresses, mother, mothers, mummy, niece, nun, nuns, priestesses, princess, princesses, prioress, queen, sister, sisters, wench, widow, widows, wife, wives, woman, women.

Neutral

admirer, baggage, baritone, bass, beast, beasts, beauty, beaver, bombshells, buddy, cousin, cows, creole, cuckoo, dean, devil, devils, dish, dishes, diva, divas, dive, dolls, dragon, dragons, drone, dwarf, dwarfs, equestrian, fairy, fellow, fellows, flirt, gob, god, ham, headmaster, hero, heroes, heros, hijackers, hogs, homunculus, housekeeper, jade, jewel, junior, knockout, knockouts, letters, lizard, louts, lover, lovers, master, member, members, metropolitan, mezzo, minor, minors, name, names, nurse, nurses, orphan, parachutists, patron, patrons, peaches, pop, punchers, punk, rams, rapists, reaper, rock, rocks, sage, sages, satyr, schoolmaster, scrubber, senior, seniors, siren, skinhead, skinheads, skirt, skirts, soprano, spectator, spectators, sphinx, stud, sweethearts, tart, tenor, thugs, virgin, youth, youths.

A.2.2 Lexicons/seed nouns used for dataset creation

Masculine

novillero, ponce, schoolboy, vaquero, casanova, indiana, sheepman, inamorato, basileus, priest, signore, tom, milord, pachuco, flamen, headsman, fireman, franklin, everyman, peer, brother, lawman, cavalryman, cleric, committeeman, boyfriend, klansman, archbishop, longbowman, quarryman, suitor, masseur, helmsman, wittol, hobbledehoy, man, ganger, milkman, bellboy, romeo, galoot, weatherman, swagman, liveryman, fathead, butler, king, sod, emir, tenor, caveman, iceman, bushman, kinsman, proconsul, cuckold, statesman, ex-boyfriend, lineman, gent, ordinary, patriarch, ethnarch, capo, infantryman, sheik, assemblyman, frontiersman, son, womanizer, guardsman, beggarman, orangeman, wolf, minuteman, countertenor, mr., burgrave, goliard, militiaman, gillie, aircrewman, schoolman, charon, town, coiffeur, sultan, placeman, husband, sannup, geezer, tarzan, loon, gaucho, foster-brother, paperboy, samurai, selectman, forefather, ape-man, repairman, housefather, dalesman, don, archdeacon, mailman, raja, mafioso, cyril, polygynist, poultryman, sandboy, imam, showman, margrave, yeoman, plainsman, businessman, dandy, sidesman, excavator, choirboy, foster-son, plainclothesman, aircraftsman, plowboy, ejaculator, warlord, outdoorsman, vestryman, mikado, beadsman, chapman, uxoricide, stepbrother, councilman, irishman, sadhu, englishman, trainman, townes,

grandfather, busboy, count, footman, great-nephew, liege, adonis, ottoman, oklahoman, oilman, monsignor, viscount, crewman, sir, satyr, bey, lord, jacob, roundsman, son-in-law, bass, godson, townsman, gentlemen, yachtsman, clansman, eparch, gentleman, brakeman, pitchman, sire, exarch, armiger, cub, centurion, father-figure, coachman, ferryman, nuncio, vicar-general, foster-father, duke, strongman, man-at-arms, brunet, workman, princeling, abbot, artilleryman, rifleman, castrato, polycarp, timberman, freedman, baritone, benedick, guy, pope, nephew, ironside, rake, stepfather, salesman, posseman, lothario, congressman, bagman, welshman, horseman, plowman, tribesman, uriah, dayboy, layman, groomsman, libertine, subdeacon, male, cameraman, trainbandsman, spokesman, kennan, babu, policeman, shah, deliveryman, earl, wireman, maharaja, cattleman, gunman, serviceman, widower, buddy, wingman, draftsman, pharaoh, lumberman, father-in-law, blade, vizier, seedsman, lobsterman, tallyman, gasman, augustinian, roadman, knight-errant, handyman, stableman, freeman, trappist, marquess, shaver, lackey, soundman, fauntleroy, emperor, signor, uncle, vicar, praetor, men, danseur, lighterman, father, mullah, manservant, alderman, fisherman, yardman, squire, palatine, catamite, fugleman, dad, gagman, stepson, kaiser, thane, doorkeeper, pater, signalman, stockman, raftsman, friar, grandson, mollycoddle, baron, boy, pointsman, marquis, washerman, groom, sodomite, czar, t-man, fellow, nabob, ex-husband, deacon, groundsman, foreman, woodsman, esquire, clergyman, coalman, boatman, macho, curandero, brother-in-law, linesman, cowboy, shaheed, bondsman, doughboy, elector, jimdandy, coastguardsman, great-uncle, muscleman, sandwichman, limey, pederast, stiff, herr, godfather, macaroni, malik, baggageman, misogynist, page, bacchant, marksman, frenchman, letterman, dairyman, bondman, dean, popper, jawan, cornishman, homeboy, dragoman, pendragon, senhor, headman, mestizo, watchman, dauphin, gitano, lowerclassman, countryman, stud, townee, pilate, midshipman, mover, oarsman, g-man, roman, jute, seducer, cavalier, bullyboy, hangman, bull, brahman, archpriest, batman, wencher, monsieur, simeon, codger, apostle, linkboy, sirrah, bachelor, craftsman, bandsman, grandee, potboy, monk

Feminine

coiffeuse, landlady, chambermaid, comedienne, odalisque, squire, countrywoman, nymphet, instructress, hag, spokeswoman, flibbertigibbet, nanny, quadripara, wac, daughter, starlet, scold, authoress, executrix, squaw, ingenue, oarswoman, parlormaid, stateswoman, jilt, forewoman, sylph, englishwoman, beard, great-aunt, heiress, ma'am, schoolgirl, cinderella, irishwoman, protegee, debutante, foster-nurse, stepmother, brownie, spinster, traitress, patroness, marchioness, groupie, cornishwoman, madwoman, beggarwoman, mayoress, mediatrix, signorina, prophetess, memsahib, bondwoman, concubine, mammy, mrs., tomboy, co-ed, begum, assemblywoman, usherette, deaconess, ladylove, gamine, gitana, mother-in-law, countess, madame, ms., primigravida, sculptress, yenta, priestess, curandera, selectwoman, vestrywoman, women, demimondaine, foster-mother, spitfire, viscountess, doyenne, mother, homegirl, ball-buster, primipara, barmaid, bride, niece, thrush, donna, godmother, hostess, coloratura, light-o'-love, manageress, czarina, damsel, postmistress, councilwoman, cowgirl, procuress, handmaid, secundigravida, proprietress, foundress, girlfriend, signora, colleen, bag, milady, broad, horsewoman, baggage, beguine, enchantress, duchess, lady, female, seductress, chachka, smasher, stepdaughter, wife, sheika, mamma, salesgirl, she-devil, geisha, granddaughter, granny, sister-in-law, heroine, gal, jewess, dragon, bacchante, canary, duenna, daygirl, negotiatress, frontierswoman, circe, nan, jezebel, foremother, benefactress, coquette, lass, niqaabi, poetess, shrew, virago, dame, undoer, housewife, maenad, stewardess, villainess, beldam, foster-sister, harridan, nullipara, fiancee, girl, charwoman, soprano, butch, princess, quintipara, housemother, bluestocking, shiksa, newswoman, maid, nun, soubrette, bridesmaid, shepherdess, frump, divorcee, madam, maharani, wardress, nymph, poseuse, committeewoman, vestal, belle, muslimah, mill-girl, skivvy, washwoman, baroness, matron, aunt, farmerette, gravida, adventuress, dairymaid, widow, waitress, uxor, adulteress, kinswoman, slattern, sorceress, conductress, sibyl, archduchess, chatelaine, woman, supermom, ladies, eyeful, abbess, millionairess, embroideress, outdoorswoman, scotswoman, empress, foster-daughter, mistress, peri, wanton, nymphomaniac, ex-wife, contralto, amazon,

great-niece, mannequin, murderess, businesswoman, idolatress, ma, queen, huntress, songstress, actress, ayah, midwife, polyandrist, schoolmarm, ambassadress, rani, goddaughter, rosebud, aviatrix, daughter-in-law, flapper, testatrix, matriarch, confidante, mestiza, suffragette, puerpera, b-girl, taskmistress, tertigravida, ballerina, governess, inamorata, ancestress, parisienne, headmistress, bobbysoxer, sister

Neutral

Male stereotype: actor, architect, ballplayer, barber, baron, bishop, bodyguard, butcher, butler, captain, custodian, dean, fighter_pilot, financier, footballer, gangster, janitor, laborer, maestro, magician, major_leaguer, mechanic, midfielder, mobster, pastor, philosopher, preacher, president, priest, soldier, sportswriter, taxi_driver, trucker, warden, warrior, welder,
Female Stereotype: dancer, hairdresser, homemaker, housekeeper, librarian, nanny, nurse, receptionist, secretary, socialite, stylist,
Neutral: accountant, administrator, advocate, aide, analyst, anthropologist, artist, artiste, associate_dean, associate_professor, attorney, author, baker, bartender, biologist, campaigner, cellist, chemist, citizen, civil_servant, collector, columnist, comedian, commentator, consultant, correspondent, councilor, counselor, curator, dermatologist, doctoral_student, employee, entertainer, envoy, environmentalist, epidemiologist, freelance_writer, gardener, geologist, graphic_designer, guidance_counselor, illustrator, instructor, journalist, jurist, lecturer, lyricist, mediator, missionary, musician, narrator, naturalist, negotiator, novelist, observer, painter, parishioner, pathologist, pediatrician, performer, pharmacist, philanthropist, photographer, photojournalist, pianist, planner, poet, pollster, promoter, prosecutor, protester, provost, psychiatrist, psychologist, publicist, pundit, radiologist, researcher, restaurateur, servant, singer, singer_songwriter, sociologist, strategist, student, swimmer, teenager, treasurer, trumpeter, tutor, undersecretary, valedictorian, vocalist, worker, writer

Appendix B

Prompts

B.1 Few-Shot Prompt Given to Qwen 2.5-3B and Qwen 2.5-72B

The prompt provided to the two Qwen models is a slightly adapted version of the prompt used by Pranav et al. (2025) to pre-label Glitter, a dataset for gender-fair machine translation from English to German. To adapt the prompt to the task, more examples from the development set were added to cover all relevant gender cues.

```
You are a data filtering assistant. Your task is to determine
whether the gender of the seed word is disambiguated by the context
(previous sentence, matching sentence, and trailing sentence). Avoid
 making assumptions based on names or roles. Include the words "
AMBIGUOUS" or "UNAMBIGUOUS" in your response. Keep your response
concise and to the point.

CLEARLY DISAMBIGUATED EXAMPLES:

Female Quality Technician Example:
Seed: "technician"
Text: "The quality technician will participate in the development of
 new products by integrating quality requirements from the design
phase. Statistical analysis methods will be used to improve
processes. Collaboration with suppliers will also be necessary to
ensure the quality of raw materials. Project teams will trust her to
 guarantee product excellence."
Output: I can see that the female pronoun 'her' is connected to the
quality technician. UNAMBIGUOUS (FEMALE).

Female Colleague Example:
Seed: "colleague"
Text: "They may also fall under the competence of my colleague, Mrs
Barnier, who spoke to this Assembly a little while ago."
Output: I can see that the female title "Mrs" refers to the
colleague. UNAMBIGUOUS (FEMALE).

Female Employees Example:
Seed: "employees"
Previous: "The Kuwait government subsidizes certain types of
finances for divorced or widowed Kuwaiti women provided they have
children to care for. The Kuwait Labour Code provides maternity
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leave for women, who can receive 70-day of paid leave and up to four
 months of optional unpaid leave, during which employment
termination is illegal."
Matching: "Additionally, employers with more than 50 female
employees must provide infant childcare facilities by law."
Trailing: "The Civil Service Committee Decree of 1993 discriminates
between women married to Kuwaiti men versus non-Kuwaiti men."
Output: I can see that the text explicitly uses the phrase "female
employees" to specify gender. UNAMBIGUOUS (FEMALE).

Female Participants Example:
Seed: "participants"
Previous: "The study focused on maternal health outcomes in rural
communities."
Matching: "All participants were mothers who had given birth within
the last year, recruited from local health centers."
Trailing: "The women reported varying levels of access to prenatal
care."
Output: Based on multiple gender indicators - "mothers" and "women"
- the participants are clearly identified as female. UNAMBIGUOUS (
FEMALE).

Lady Example:
Seed: "lady"
Text: '"There is no knowing what THEY may expect," said the lady, "
but we are not to think of their expectations: the question is, what
 you can afford to do.'
Output: The word 'lady' always refers to a woman. UNAMBIGUOUS (
FEMALE).

Male Joiner Example:
Seed: "joiner"
Text: "The joiner designed a carved front door for a historic
building. His preliminary drawings were approved by the architect
before he began the work."
Output: The male pronoun "His" refers to the joiner. UNAMBIGUOUS (
MALE).

Male Minister Example:
Seed: "minister"
Text: "I heard, and it is not an isolated incident, the United
Kingdom Minister for Social Welfare say on television that if he
could stop fraud he could save GBP 6 billion."
Output: I can see that the male pronoun 'he' refers to the word
Minister. UNAMBIGUOUS (MALE).

Male Competitors Example:
Seed: "competitors"
Previous: "The heavyweight boxing division saw record attendance."
Matching: "The competitors, all male boxers between ages 20-35, were
 required to meet strict weight requirements."
Trailing: "Each fighter underwent medical screening before the
matches."
Output: I can see that the text explicitly specifies "male boxers"
when referring to the competitors. UNAMBIGUOUS (MALE).
```

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Husband Example:
Seed: "husband"
Text: "I'm honey only to my husband, understand??"
Output: The word 'husband' is masculine and always refers to a man.
UNAMBIGUOUS (MALE).

Parents as Trainers Example:
Seed: "trainers"
Previous: "Irina Iosifovna Turova (; born 10 August 1979), Irina
Slavina (, also Irina Slavina–Turova), is a Russian chess player who
 holds the titles of Woman Grandmaster (2001) and International
Master (2004). She won the Russian Women's Chess Championship in
2003."
Matching: "Biography Irina was born in Belarus, where her parents
worked as chess trainers."
Trailing: "After Chernobyl disaster, her family moved to Arkhangelsk
, where at age of eight Irina won the second place in the
Arkhangelsk Oblast Women's Chess Championship."
Output: Based on the context, the word "parents" clearly indicates
both male and female trainers working together. UNAMBIGUOUS (BOTH).

Explicit Gender Mix Example:
Seed: "respondents"
Previous: "In Russia, according to a 2011 survey by the Southern
Federal University, brunettes are considered more attractive than
blondes."
Matching: "It is important to note that among the respondents in
this study were 50% men and 50% women."
Trailing: "Another study by the University of Tampa, which also used
 male and female students..."
Output: Looking at the matching sentence, there's an explicit split
of "50% men and 50% women" among the respondents. UNAMBIGUOUS (BOTH)
.

Research Team Example:
Seed: "researchers"
Previous: "The international collaboration spanned multiple
universities."
Matching: "The research team consisted of an equal split of male and
 female researchers, with 15 men and 15 women from various
scientific backgrounds."
Trailing: "Their diverse perspectives contributed to the study's
comprehensive approach."
Output: Looking at the matching sentence, there's an explicit count
of "15 men and 15 women" researchers. UNAMBIGUOUS (BOTH).

AMBIGUOUS EXAMPLES:

Physiotherapist Example:
Seed: "physiotherapist"
Text: "This physiotherapist designed a tailored exercise plan. They
gained attention for their personalized approach."
Output: Based on the context, the singular pronoun "they" signals
that the physiotherapist's gender is ambiguous. AMBIGUOUS.
```

Doctor Example:
Seed: "doctor"
Text: "The doctor prescribes additional tests to refine the diagnosis."
Output: There are no pronouns or other cues that could tell me the doctor's gender. AMBIGUOUS.

European Citizens Example:
Seed: "citizens"
Text: "The Commission needs to draw up a list of the problems encountered by European citizens, particularly with regard to the issues of freedom of residence and freedom of establishment and, most especially, with regard to their right to work, study and obtain recognition of their qualifications in other Member States."
Output: There are no titles, pronouns, or cues that could point towards the citizens' gender. AMBIGUOUS.

Lawyers Example:
Seed: "lawyers"
Previous: "Earlier it was only given on certain grounds. Within the first two years of passing this law, the courts saw an exponential increase in khul' lawsuits."
Matching: "The law has yet to be approved by parliament, however, and it is still condemned by many lawyers to this day."
Trailing: "Nigeria Khul' is the most common form of divorce in Northern Nigeria."
Output: I see no gender indicators or specifications for these lawyers in any of the context sentences. AMBIGUOUS.

Investigators Example:
Seed: "investigators"
Previous: "A total of 43 knife wounds were inflicted on Uemura's body, including 31 to the neck during the assault, which lasted more than an hour. Populist weekly Shukan Shincho reported the wounds appeared to indicate that whoever killed Ryota may have been trying to decapitate him."
Matching: ""Some investigators suspect (the criminals) watched Internet videos showing the execution of hostages by Islamic State (IS) fighters and sought to mimic them," the magazine said, quoting an unnamed source close to police."
Trailing: "Prosecution On 27 February 2015, an 18-year-old boy was arrested by Kanagawa Police on suspicion of murder."
Output: Based on the context, there are no gender markers or specifications for the investigators mentioned. AMBIGUOUS.

Economists Example:
Seed: "economists"
Previous: "In response, the Club sponsored a petition of 1,028 economists who stated their opposition to protectionist policies against China. The list of economists included Nobel Laureates Finn Kydland, Edward Prescott, Thomas Schelling, and Vernon Smith."
Matching: "The petition played off a similar petition that was also signed by 1,028 economists in 1930 that opposed the Smoot–Hawley Tariff Act."

```
Trailing: "In 2008 and 2009, the Club for Growth opposed the $787
billion stimulus bill, Cash for Clunkers, cap and trade legislation,
 the Wall Street bailout, the auto bailout, the Affordable Care Act
and the bailout of Fannie Mae and Freddie Mac."
Output: Looking at all context segments, I find no gender indicators
 for the general group of economists. AMBIGUOUS.

Beginners Example:
Seed: "beginners"
Previous: "Their game entertains without annoying, and there aren't
many games, especially by mail, of which that can be said." In 1986,
 reviewer Dale A. Perkins stated that "If you are into Dungeons and
Dragons' style combat, this is the game for you", recommending
trying the game, regardless of gaming background."
Matching: "In 1991, reviewer Vickie Lloyd advised that her concerns
with the game were "very minor" and Quest was "a great game and I
very much recommend it", especially for beginners."
Trailing: "In 1985, the game tied with DuelMasters, Pellic Quest,
and Power for Third Place in the 1st Annual Paper Mayhem Awards for
"Best PBM Game"."
Output: I see no gender markers or specifications for the beginners
mentioned in any of the context sentences. AMBIGUOUS.

Artist Example:
Seed: "artist"
Text: "An anarchist is an artist."
Output: There are no titles, pronouns, or cues that could point
towards artist's gender. Artist itself is a neutral word. AMBIGUOUS.
```

B.2 Prompt given to Qwen 2.5-3B for the LLM-Extension of the rule-based system

```
You are a helpful assistant.
A gender cue is a word that can give us information about the gender
 of a noun.
Your task is to decide whether a given gender cue influences the
gender of a specific word in the text passage.
Reply with one word!

Example 1:
Passage: "My mother preferred to go to [female] <<doctors>>."
Question: Does the gender cue [female] influence the gender of <<<
doctors>>>?
Here, you should reply with 'yes' since the gender cue [female]
tells us that the doctors are female.

Example 2:
Passage: "Early <<settlers>> played pivotal roles in building a
hospital for [women]."
Question: Does the gender cue [women] influence the gender of <<<
settlers>>>?
Here, you should reply with 'no' since the gender cue [women] refers
 to patients of the hospital and not the setllers.
```

```
Example 3:
Passage: "The [man] who had known my father since they were children
 was a great <<friend>>."
Question: Does the gender cue [man] influence the gender of <<<
friend>>>?
Here, you should reply with 'yes' since the gender cue 'man' refers
to the same person as 'friend'.

Example 6:
Passage: "<<Interpreters>> are often found bridging gaps in
international summits. The expertise these [men] have in multiple
languages is indispensable."
Question: Does the gender cue [men] influence the gender of <<<
interpreters>>>?
Here, you should reply with 'yes', since the gender cue 'men' refers
 to the interpreters, meaning they are male.
```

Appendix C

Full Result Tables

Table C.1: FairTranslate Full Results (Precision, Recall, F1, Accuracy)

	ambiguous ($n = 259$)			unamb. female ($n = 139$)			unamb. male ($n = 142$)				macro avg			**weighted avg**		
System	P	R	F_1	P	R	F_1	P	R	F_1	**acc**	P	R	F_1	P	R	F_1
Majority	0.480	1.000	0.648	0.000	0.000	0.000	0.000	0.000	0.000	0.480	0.160	0.333	0.216	0.230	0.480	0.311
Random	0.479	0.313	0.379	0.241	0.324	0.276	0.255	0.331	0.288	0.320	0.325	0.322	0.314	0.359	0.320	0.328
Rule	0.944	0.985	0.964	0.992	0.935	0.963	0.978	0.958	0.968	0.965	0.972	0.959	0.965	0.966	0.965	0.965
LogReg	0.938	0.988	0.962	0.992	0.928	0.959	0.985	0.951	0.968	0.963	0.972	0.956	0.963	0.964	0.963	0.963
DistilBERT	**1.000**	**0.996**	**0.998**	**0.993**	**1.000**	**0.996**	**1.000**	**1.000**	**1.000**	**0.998**	**0.998**	**0.999**	**0.998**	**0.998**	**0.998**	**0.998**
Qwen-3B	0.779	0.819	0.798	0.829	0.626	0.713	0.652	0.739	0.693	0.748	0.565	0.546	0.551	0.759	0.748	0.749
Qwen-72B	0.989	0.996	0.992	0.993	0.993	0.993	1.000	0.986	0.993	0.993	0.994	0.992	0.993	0.993	0.993	0.993
Rule+LLM	0.944	0.985	0.964	0.992	0.935	0.963	0.978	0.958	0.968	0.965	0.972	0.959	0.965	0.966	0.965	0.965

Table C.2: GeNTE Full Results (Precision, Recall, F1, Accuracy)

	ambiguous ($n = 60$)			unamb. female ($n = 94$)			unamb. male ($n = 88$)				macro avg			**weighted avg**		
System	P	R	F_1	P	R	F_1	P	R	F_1	**acc**	P	R	F_1	P	R	F_1
Majority	0.000	0.000	0.000	0.388	1.000	0.560	0.000	0.000	0.000	0.388	0.129	0.333	0.187	0.151	0.388	0.217
Random	0.227	0.283	0.252	0.356	0.340	0.348	0.325	0.284	0.303	0.306	0.302	0.303	0.301	0.312	0.306	0.308
Rule	0.488	0.667	0.563	1.000	0.777	0.874	0.770	0.761	0.766	0.744	0.753	0.735	0.734	0.789	0.744	0.758
LogReg	0.494	0.683	0.573	1.000	0.766	0.867	0.782	0.773	0.777	0.748	0.759	0.741	0.739	0.795	0.748	0.762
DistilBERT	**1.000**	**0.950**	**0.974**	**1.000**	**1.000**	**1.000**	**0.967**	**1.000**	**0.983**	**0.988**	**0.989**	**0.983**	**0.986**	**0.988**	**0.988**	**0.988**
Qwen-3B	0.397	0.967	0.563	0.919	0.606	0.731	0.562	0.205	0.300	0.550	0.470	0.444	0.398	0.660	0.550	0.533
Qwen-72B	0.857	0.900	0.878	1.000	0.989	0.995	0.930	0.909	0.920	0.938	0.929	0.933	0.931	0.939	0.938	0.938
Rule+LLM	0.488	0.667	0.563	1.000	0.777	0.874	0.770	0.761	0.766	0.744	0.753	0.735	0.734	0.789	0.744	0.758

Table C.3: Glitter Full Results (Precision, Recall, F1, Accuracy)

	ambiguous ($n = 205$)			unamb. all ($n = 32$)			unamb. female ($n = 79$)			unamb. male ($n = 76$)				macro avg			**weighted avg**		
System	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1	**acc**	P	R	F_1	P	R	F_1
Majority	0.523	1.000	0.687	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.523	0.131	0.250	0.172	0.273	0.523	0.359
Random	0.495	0.224	0.309	0.058	0.188	0.089	0.210	0.266	0.235	0.177	0.224	0.198	0.230	0.235	0.225	0.207	0.340	0.230	0.254
Rule	0.597	0.946	0.732	0.000	0.000	0.000	0.816	0.392	0.530	0.917	0.289	0.440	0.630	0.582	0.407	0.425	0.654	0.630	0.575
LogReg	0.626	0.922	0.746	0.000	0.000	0.000	0.821	0.405	0.542	0.608	0.408	0.488	0.643	0.514	0.434	0.444	0.610	0.643	0.594
DistilBERT	**0.811**	**0.849**	**0.829**	**0.820**	**0.448**	**0.578**	0.733	0.810	0.769	**0.784**	**0.746**	**0.764**	**0.788**	**0.787**	**0.713**	**0.735**	**0.791**	**0.788**	**0.784**
Qwen-3B	0.527	0.912	0.668	0.250	0.062	0.100	0.545	0.152	0.238	0.143	0.013	0.024	0.515	0.366	0.285	0.257	0.434	0.515	0.410
Qwen-72B	0.719	0.873	0.789	0.520	0.406	0.456	**0.827**	**0.785**	**0.805**	0.907	0.513	0.655	0.747	0.743	0.644	0.676	0.761	0.747	0.739
Rule+LLM	0.626	0.943	0.752	0.000	0.000	0.000	0.823	0.452	0.583	0.931	0.412	0.572	0.664	0.595	0.452	0.477	0.674	0.664	0.622

Table C.4: Lexical Gender Extension Full Results (Precision, Recall, F1, Accuracy)

	ambiguous ($n = 127$)			unamb. female ($n = 152$)			unamb. male ($n = 142$)				macro avg†			**weighted avg**		
System	P	R	F_1	P	R	F_1	P	R	F_1	**acc**	P	R	F_1	P	R	F_1
Majority	0.000	0.000	0.000	0.361	1.000	0.531	0.000	0.000	0.000	0.361	0.120	0.333	0.177	0.130	0.361	0.192
Random	0.336	0.346	0.341	0.384	0.382	0.383	0.388	0.380	0.384	0.371	0.369	0.369	0.369	0.371	0.371	0.371
Rule	**0.836**	**1.000**	**0.910**	**1.000**	**0.974**	**0.987**	**0.992**	**0.845**	**0.913**	**0.938**	**0.942**	**0.940**	**0.937**	**0.948**	**0.938**	**0.939**
LogReg	0.818	0.992	0.897	0.993	0.974	0.983	0.983	0.817	0.892	0.926	0.932	0.928	0.924	0.937	0.926	0.927
DistilBERT	0.882	0.958	0.918	0.958	0.945	0.951	0.965	0.903	0.933	0.935	0.935	0.936	0.935	0.937	0.935	0.935
Qwen-3B	0.365	0.945	0.526	0.440	0.145	0.218	0.436	0.120	0.188	0.378	$0.310^{\dagger}$	$0.302^{\dagger}$	$0.233^{\dagger}$	0.416	0.378	0.301
Qwen-72B	0.651	0.984	0.784	0.984	0.836	0.904	0.989	0.662	0.793	0.822	0.656	0.620	0.620	0.886	0.822	0.830
Rule+LLM	**0.836**	**1.000**	**0.910**	**1.000**	**0.974**	**0.987**	**0.992**	**0.845**	**0.913**	**0.938**	**0.942**	**0.940**	**0.937**	**0.948**	**0.938**	**0.939**

Table C.5: Full Overall Results (Precision, Recall, F1, Accuracy)

	ambiguous ($n = 651$)			unamb. all ($n = 32$)			unamb. female ($n = 464$)			unamb. male ($n = 448$)				macro avg			**weighted avg**		
System	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1	**acc**	P	R	F_1	P	R	F_1
Majority	0.408	1.000	0.580	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.408	0.102	0.250	0.145	0.167	0.408	0.237
Random	0.427	0.264	0.326	0.015	0.188	0.028	0.318	0.265	0.289	0.303	0.272	0.287	0.265	0.266	0.247	0.232	0.352	0.265	0.298
Rule	0.743	0.946	0.832	0.000	0.000	0.000	0.979	0.823	0.895	0.930	0.770	0.842	0.842	0.663	0.635	0.642	0.849	0.842	0.837
LogReg	0.754	0.940	0.837	0.000	0.000	0.000	0.977	0.821	0.892	0.891	0.781	0.832	0.842	0.655	0.636	0.640	0.842	0.842	0.835
DistilBERT	**0.915**	**0.938**	**0.926**	**0.820**	**0.448**	**0.578**	**0.935**	**0.950**	**0.942**	**0.947**	**0.927**	**0.936**	**0.928**	**0.904**	**0.816**	**0.846**	**0.928**	**0.928**	**0.927**
Qwen-3B	0.524	0.886	0.658	0.133	0.062	0.085	0.745	0.384	0.506	0.590	0.315	0.410	0.563	0.498	0.412	0.415	0.599	0.563	0.533
Qwen-72B	0.805	0.946	0.870	0.433	0.406	0.419	0.963	0.905	0.933	0.970	0.788	0.869	0.879	0.793	0.761	0.773	0.890	0.879	0.879
Rule+LLM	0.757	0.945	0.840	0.000	0.000	0.000	0.978	0.833	0.900	0.931	0.791	0.855	0.850	0.666	0.642	0.649	0.855	0.850	0.845

Table C.6: DistilBERT classification results (P, R, F_1) per dataset and class for 3 seeds; arithmetic mean ± standard deviation

			Seed 14			**Seed 45**			**Seed 98**			**Final**		
Dataset	**Class**	n	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1
FairTranslate ($n = 540$)	ambiguous	259	1.000	0.996	0.998	1.000	0.996	0.998	1.000	0.996	0.998	1.000	0.996	0.998 ± 0.000
	unambig. female	139	0.993	1.000	0.996	0.993	1.000	0.996	0.993	1.000	0.996	0.993	1.000	0.996 ± 0.000
	unambig. male	142	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000 ± 0.000
	accuracy	540	—	—	*0.998*	—	—	*0.998*	—	—	*0.998*	—	—	0.998 ± 0.000
	macro avg	540	0.998	0.999	0.998	0.998	0.999	0.998	0.998	0.999	0.998	0.998	0.999	0.998 ± 0.000
	weighted avg	540	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998 ± 0.000
GeNTE ($n = 242$)	ambiguous	60	1.000	0.950	0.974	1.000	0.950	0.974	1.000	0.950	0.974	1.000	0.950	0.974 ± 0.000
	unambig. female	94	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000 ± 0.000
	unambig. male	88	0.967	1.000	0.983	0.967	1.000	0.983	0.967	1.000	0.983	0.967	1.000	0.983 ± 0.000
	accuracy	242	—	—	*0.988*	—	—	*0.988*	—	—	*0.988*	—	—	0.988 ± 0.000
	macro avg	242	0.989	0.983	0.986	0.989	0.983	0.986	0.989	0.983	0.986	0.989	0.983	0.986 ± 0.000
	weighted avg	242	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988 ± 0.000
Glitter ($n = 392$)	ambiguous	205	0.805	0.844	0.824	0.825	0.829	0.827	**0.803**	**0.873**	**0.836**	0.811	0.849	0.829 ± 0.006
	unambig. all	32	0.737	0.438	0.549	0.789	0.469	0.588	**0.933**	**0.438**	**0.596**	0.820	0.448	0.578 ± 0.025
	unambig. female	79	0.741	0.797	0.768	**0.717**	**0.835**	**0.772**	0.741	0.797	0.768	0.733	0.810	0.769 ± 0.002
	unambig. male	76	0.767	0.737	0.752	0.773	0.763	0.768	**0.812**	**0.737**	**0.772**	0.784	0.746	0.764 ± 0.011
	accuracy	392	—	—	*0.781*	—	—	*0.788*	—	—	***0.796***	—	—	0.788 ± 0.008
	macro avg	392	0.762	0.704	0.723	0.776	0.724	0.739	**0.822**	**0.711**	**0.743**	0.787	0.713	0.735 ± 0.011
	weighted avg	392	0.779	0.781	0.776	0.791	0.788	0.785	**0.803**	**0.796**	**0.791**	0.791	0.788	0.784 ± 0.008
Lexical ($n = 421$)	ambiguous	127	0.864	0.953	0.906	**0.891**	**0.961**	**0.924**	**0.891**	**0.961**	**0.924**	0.882	0.958	0.918 ± 0.010
	unambig. female	152	0.954	0.947	0.950	**0.960**	**0.947**	**0.954**	0.960	0.941	0.950	0.958	0.945	0.951 ± 0.002
	unambig. male	142	0.969	0.887	0.926	0.963	0.908	0.935	**0.963**	**0.915**	**0.939**	0.965	0.903	0.933 ± 0.007
	accuracy	421	—	—	*0.929*	—	—	***0.938***	—	—	***0.938***	—	—	0.935 ± 0.005
	macro avg	421	0.929	0.929	0.928	**0.938**	**0.939**	**0.938**	**0.938**	**0.939**	**0.938**	0.935	0.936	0.935 ± 0.006
	weighted avg	421	0.932	0.929	0.929	**0.940**	**0.938**	**0.938**	**0.940**	**0.938**	**0.938**	0.937	0.935	0.935 ± 0.005
Overall ($n = 1595$)	ambiguous	651	0.909	0.935	0.922	0.922	0.932	0.927	**0.913**	**0.946**	**0.929**	0.915	0.938	0.926 ± 0.004
	unambig. all	32	0.737	0.438	0.549	0.789	0.469	0.588	**0.933**	**0.438**	**0.596**	0.820	0.448	0.578 ± 0.025
	unambig. female	464	0.936	0.948	0.942	**0.931**	**0.955**	**0.943**	0.938	0.946	0.942	0.935	0.950	0.942 ± 0.001
	unambig. male	448	0.945	0.920	0.932	0.943	0.931	0.937	**0.952**	**0.929**	**0.940**	0.947	0.927	0.936 ± 0.004
	accuracy	1595	—	—	*0.925*	—	—	*0.929*	—	—	***0.931***	—	—	0.928 ± 0.003
	macro avg	1595	0.882	0.810	0.836	0.897	0.822	0.849	**0.934**	**0.815**	**0.852**	0.904	0.816	0.846 ± 0.009
	weighted avg	1595	0.924	0.925	0.923	0.928	0.929	0.928	**0.931**	**0.931**	**0.929**	0.928	0.928	0.927 ± 0.003

Table C.7: Rule-Based + LLM (Qwen 2.5-3B Instruct) classification results (P, R, F_1) per dataset and class for 3 seeds; arithmetic mean ± standard deviation

			Seed 14			**Seed 48**			**Seed 98**			**Final**		
Dataset	**Class**	n	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1
FairTranslate ($n = 540$)	ambiguous	259	0.944	0.985	0.964	0.944	0.985	0.964	0.944	0.985	0.964	0.944	0.985	0.964 ± 0.000
	unambig. female	139	0.992	0.935	0.963	0.992	0.935	0.963	0.992	0.935	0.963	0.992	0.935	0.963 ± 0.000
	unambig. male	142	0.978	0.958	0.968	0.978	0.958	0.968	0.978	0.958	0.968	0.978	0.958	0.968 ± 0.000
	accuracy	540	—	—	*0.965*	—	—	*0.965*	—	—	*0.965*	—	—	0.965 ± 0.000
	macro avg	540	0.972	0.959	0.965	0.972	0.959	0.965	0.972	0.959	0.965	0.972	0.959	0.965 ± 0.000
	weighted avg	540	0.966	0.965	0.965	0.966	0.965	0.965	0.966	0.965	0.965	0.966	0.965	0.965 ± 0.000
GeNTE ($n = 242$)	ambiguous	60	0.488	0.667	0.563	0.488	0.667	0.563	0.488	0.667	0.563	0.488	0.667	0.563 ± 0.000
	unambig. female	94	1.000	0.777	0.874	1.000	0.777	0.874	1.000	0.777	0.874	1.000	0.777	0.874 ± 0.000
	unambig. male	88	0.770	0.761	0.766	0.770	0.761	0.766	0.770	0.761	0.766	0.770	0.761	0.766 ± 0.000
	accuracy	242	—	—	*0.744*	—	—	*0.744*	—	—	*0.744*	—	—	0.744 ± 0.000
	macro avg	242	0.753	0.735	0.734	0.753	0.735	0.734	0.753	0.735	0.734	0.753	0.735	0.734 ± 0.000
	weighted avg	242	0.789	0.744	0.758	0.789	0.744	0.758	0.789	0.744	0.758	0.789	0.744	0.758 ± 0.000
Glitter ($n = 392$)	ambiguous	205	0.625	0.941	0.751	**0.629**	**0.941**	**0.754**	0.624	0.946	0.752	0.626	0.943	0.752 ± 0.002
	unambig. all	32	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000 ± 0.000
	unambig. female	79	0.814	0.443	0.574	0.818	0.456	0.585	**0.837**	**0.456**	**0.590**	0.823	0.452	0.583 ± 0.008
	unambig. male	76	**0.941**	**0.421**	**0.582**	0.914	0.421	0.577	0.938	0.395	0.556	0.931	0.412	0.572 ± 0.014
	accuracy	392	—	—	*0.663*	—	—	***0.666***	—	—	*0.663*	—	—	0.664 ± 0.002
	macro avg	392	0.595	0.451	0.477	**0.590**	**0.455**	**0.479**	0.600	0.449	0.474	0.595	0.452	0.477 ± 0.003
	weighted avg	392	0.673	0.663	0.621	**0.671**	**0.666**	**0.624**	0.677	0.663	0.620	0.674	0.664	0.622 ± 0.002
Lexical ($n = 421$)	ambiguous	127	0.836	1.000	0.910	0.836	1.000	0.910	0.836	1.000	0.910	0.836	1.000	0.910 ± 0.000
	unambig. female	152	1.000	0.974	0.987	1.000	0.974	0.987	1.000	0.974	0.987	1.000	0.974	0.987 ± 0.000
	unambig. male	142	0.992	0.845	0.913	0.992	0.845	0.913	0.992	0.845	0.913	0.992	0.845	0.913 ± 0.000
	accuracy	421	—	—	*0.938*	—	—	*0.938*	—	—	*0.938*	—	—	0.938 ± 0.000
	macro avg	421	0.942	0.940	0.937	0.942	0.940	0.937	0.942	0.940	0.937	0.942	0.940	0.937 ± 0.000
	weighted avg	421	0.948	0.938	0.939	0.948	0.938	0.939	0.948	0.938	0.939	0.948	0.938	0.939 ± 0.000
Overall ($n = 1595$)	ambiguous	651	0.756	0.945	0.840	**0.758**	**0.945**	**0.841**	0.756	0.946	0.840	0.757	0.945	0.840 ± 0.001
	unambig. all	32	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000 ± 0.000
	unambig. female	464	0.977	0.832	0.899	0.977	0.834	0.900	**0.980**	**0.834**	**0.901**	0.978	0.833	0.900 ± 0.001
	unambig. male	448	**0.932**	**0.792**	**0.856**	0.929	0.792	0.855	0.931	0.788	0.854	0.931	0.791	0.855 ± 0.001
	accuracy	1595	—	—	*0.850*	—	—	***0.851***	—	—	*0.850*	—	—	0.850 ± 0.001
	macro avg	1595	0.666	0.642	0.649	0.666	0.643	0.649	0.667	0.642	0.649	0.666	0.642	0.649 ± 0.000
	weighted avg	1595	0.855	0.850	0.845	0.855	0.851	0.845	0.855	0.850	0.845	0.855	0.850	0.845 ± 0.000

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